

Baumslag–Solitar-free one-relator groups are hyperbolic

OpenAI

September 25, 2026

Abstract

We prove Gersten’s conjecture: every finitely generated one-relator group containing no subgroup isomorphic to a Baumslag–Solitar group $BS(m, n)$, for nonzero integers m, n , is word-hyperbolic.

Contents

1	Introduction	2
1.1	Earlier work and the remaining step	3
1.2	The combination argument	3
1.3	Magnus subgraphs	4
1.4	Structure of the proof	5
2	Intersections, cyclic commensurators, and rank bounds	5
2.1	Quasiconvex intersections	5
2.2	Commensuration in a tree action	6
2.3	Strong inertia in free groups	8
3	Annuli and currents for a restricted HNN extension	8
3.1	Frames and stabilizers of paths	9
3.2	The algebraic annular-flaring criterion	11
3.3	Currents and induction along an injection	12
3.4	From based words to currents	14
3.5	Finite transition types and a bilateral limit	16
4	A stationary current and its line trajectories	18
4.1	Periodic axes and a finite set of exceptional roots	18
4.2	A stationary current	20
4.3	Transfers on a conull set of lines	21
4.4	Actual paths and finite windows	23
5	Finite carriers for a stationary current	25
5.1	Window cores and their limits	25
5.2	A finite cofinal family of subgroups	27
5.3	Return automorphisms	29
5.4	Expansion and vanishing carrier mass	30

6	Expansion and periodic lines in the support	32
6.1	Uniform geometric estimates	33
6.2	A failed return produces a periodic segment	35
6.3	Pulling back whole periods	37
6.4	Exclusion of periodic support lines	38
6.5	The trajectory construction on the entire support	40
7	Special rays and the end of the combination proof	41
7.1	Finitely many rays with two pasts	42
7.2	A return map with positive lag	42
7.3	A cylinder that cannot lose mass	43
8	The Magnus induction	44
8.1	Graph presentations and Magnus subgroups	45
8.2	A finite domain in a cyclic cover	45
8.3	Hyperbolicity and the final reductions	49
9	Virtual free-by-cyclic consequence	50
	References	51

1 Introduction

A single defining relation can produce very different geometries. The Baumslag–Solitar groups

$$\mathrm{BS}(m, n) = \langle a, t \mid ta^mt^{-1} = a^n \rangle, \quad m, n \in \mathbb{Z} \setminus \{0\},$$

are the basic obstructions to hyperbolicity among one-relator groups. Gersten asked whether they are the only subgroup obstructions [6, p. 228]. We prove the following statement, resolving Gersten’s conjecture positively. Here $F(X)$ denotes the free group on X , and $\langle X \mid r \rangle$ is its quotient by the normal closure of r .

Theorem 1.1. *Let X be a finite set and let $r \in F(X)$. If the one-relator group $G = \langle X \mid r \rangle$ contains no subgroup isomorphic to $\mathrm{BS}(m, n)$ for any nonzero integers m, n , then G is word-hyperbolic.*

The parameters in this statement may have either sign. In particular, the hypothesis excludes $\mathrm{BS}(1, 1) = \mathbb{Z}^2$. We call a group with this exclusion *Baumslag–Solitar-free*. The customary formulation excluding every $\mathrm{BS}(1, n)$, for every nonzero integer n , is equivalent: every Baumslag–Solitar group contains one of these groups [5, Lemma 2.10]. Theorem 1.1 includes trivial relators and proper powers; it has no torsion-free hypothesis.

We use *free-by-cyclic* in the subgroup-closed sense: a group H fits into an exact sequence

$$1 \longrightarrow K \longrightarrow H \longrightarrow Q \longrightarrow 1$$

with K free of possibly infinite rank and $Q \leq \mathbb{Z}$, possibly trivial. Kielak–Linton recall Wise’s Conjecture 17.8 that every hyperbolic one-relator group is virtually free-by-cyclic and prove the conclusion under the additional hypothesis of virtual compact specialness [10, Section 1.1 and Corollary 1.3]. The companion theorem [21, Theorem 1.1] provides that hypothesis. Combining it with Theorem 1.1 gives the Baumslag–Solitar-free form in Corollary 9.1, together with a finite-index embedding into a free-by- $\mathbb{Z}$ group with finitely generated free kernel.

1.1 Earlier work and the remaining step

The classical theory already separates the torsion case from the main difficulty. Magnus’s Freiheitssatz identifies free subgroups generated by suitable proper subsets of a one-relator generating set [17]. Newman’s spelling theorem for proper-power relators gives a Dehn algorithm and hence hyperbolicity [20, Theorem 3]; see also the proof in [18, Theorem 4]. Thus the new hyperbolicity assertion concerns non-power relators, although Theorem 1.1 includes all relators. The Freiheitssatz makes an inductive approach possible: a Magnus splitting has a smaller one-relator vertex group and free edge groups. The geometric form of the Magnus–Moldavanskii–Masters hierarchy, developed by Linton, realizes these splittings through domains in cyclic covers [12, Sections 1.1–1.2 and 4]. We use that construction in Section 8, with the domain and complexity arguments given in the form needed here.

Several recent results identify substantial classes where this induction preserves hyperbolicity. Louder and Wilton characterized negative immersions for one-relator presentation complexes by primitivity rank greater than two [15, Theorem 1.3]. Here the primitivity rank of a relator r is the least rank of a subgroup of $F(X)$ containing r in which r is not part of a free basis, with value ∞ if no such subgroup exists. Their subgroup theorem excludes Baumslag–Solitar subgroups in this class, and their subsequent work establishes uniform negative immersions and coherence [15, Corollary 1.8] [16, Theorems A and C]. Linton then proved hyperbolicity and virtual specialness for groups with negative immersions [12, Theorem 7.2]. His later characterization says that a finitely generated one-relator group is both hyperbolic and locally quasiconvex precisely when its relator has primitivity rank different from two [11, Theorem 1.4]. Local quasiconvexity means that every finitely generated subgroup is quasiconvex. This characterization does not settle hyperbolicity alone for Baumslag–Solitar-free groups whose relator has primitivity rank two.

Linton’s hierarchy criterion makes the remaining obstruction more specific: a one-relator hierarchy is quasiconvex with hyperbolic ambient group if and only if it is $\mathbb{Z}$ -stable and the group is Baumslag–Solitar-free [12, Theorem 7.1]. The stability hypothesis controls the persistence of noncyclic intersections along the hierarchy; it is an additional condition, not a consequence supplied by that criterion. A complementary reduction to primitive-extension groups isolates a particular class of two-generator one-relator groups in which to settle the conjecture [13, Theorem 5.6 and Corollary 5.7]. For a broader historical account, see [14].

The geometric issue is expansion in a graph of groups. The Bestvina–Feighn combination theorem makes annular flaring a sufficient condition for hyperbolicity when the vertex and edge data satisfy the required geometric hypotheses [1, Theorem 1.2]. We use the algebraic annular formulation in [8, Theorem 4.5]. For mapping tori of free-group automorphisms, Brinkmann proved that the absence of periodic conjugacy classes forces expansion and hyperbolicity [3]. Mutanguha extended the hyperbolicity criterion to ascending HNN extensions of finite-rank free groups: they are hyperbolic exactly when they contain no Baumslag–Solitar subgroup [19, Theorem 6.7]. His expanding relative immersions, iterated pullbacks, and annular arguments supply important precedents for treating persistent intersections [19, Theorem 4.5 and Sections 5–6]. The combination theorem here permits both free edge subgroups to be proper in a hyperbolic vertex group, under the intersection and rank hypotheses below.

1.2 The combination argument

Our main technical result is a combination theorem for a restricted class of HNN extensions of torsion-free hyperbolic groups (Theorem 3.1). The Magnus construction reduces the problem to a one-relator vertex group of smaller complexity and finitely generated free edge groups. Once the

vertex group is hyperbolic, Linton’s quasiconvexity theorem gives quasiconvex edge groups. Two further intersection properties are needed: conjugates of the edge groups have noncyclic intersection only in one specified double coset, and their overlap subgroups satisfy a strong reduced-rank inequality. Collins’s intersection theorem and Linton’s strong-inertia result supply these properties for the Magnus splittings used here [4, 12]. The combination theorem shows that the absence of Baumslag–Solitar subgroups forces annular flaring under precisely these hypotheses.

We prove flaring by contradiction, using free-group currents: invariant measures on pairs of distinct boundary points [9]. Long nonflaring annuli give a nonzero stationary current for which almost every line can be transferred across successive edges of the Bass–Serre tree. The stabilizers of increasingly long tree paths are represented by finite labeled core graphs. The rank inequality bounds the number of edges after degree-two vertices are suppressed; it does not bound their label lengths. Along a subsequence, the edges with bounded labels form one fixed graph, called the small subgraph. Its noncyclic components determine finitely many maximal subgroup carriers for persistent path stabilizers. A return along a carrier induces an atoroidal automorphism, with no nontrivial periodic conjugacy class. Brinkmann’s expansion theorem then forces the set of carrier lines to have zero current mass.

Zero mass does not exclude a carrier line from the closed support of the current. This distinction is the main issue in Section 6. Starting with a putative carrier line in the support, we transport a positive-mass cylinder along expanding carrier returns. If this transport continues to follow the prescribed carrier, recentering and compactness concentrate positive mass on its zero-mass line set. If it fails, the stabilizers of the competing actual tree path and the prescribed path have at most cyclic intersection. A sufficiently long overlap consequently follows a periodic axis of uniformly bounded primitive period. Recentering these overlaps puts a periodic line in the support. We reuse this same failure mechanism to exclude periodic support lines: pulling the overlap back along the carrier return preserves enough whole periods to identify the failure axis with the original periodic trajectory, contradicting the competing path choices. Counting whole periods is essential because inverse automorphisms can enlarge primitive period lengths.

It follows that no support line lies in the small subgraph and that every support line has irrational endpoints. The bounded graph complexity now leaves only finitely many rays admitting two different pasts. Transfer permutes their finitely many endpoint orbits under the free edge group. A return at one of these endpoints supplies infinitely many disjoint translates of a fixed positive-mass set inside a compact set, contradicting local finiteness of the current. This proves the restricted combination theorem; the geometric Magnus induction then proves Theorem 1.1.

1.3 Magnus subgraphs

We also record the geometric subgroup conclusion in the scope needed by the induction. A *circuit* in a graph is a map from a subdivided circle; it is *immersed* when its edge path has no backtracking, including at the closing point. A nontrivial element is a *non-power* if it is not u^k with $|k| \geq 2$.

Theorem 1.2. *Let Γ be a finite connected graph, and let $\lambda : S^1 \rightarrow \Gamma$ be a nontrivial immersed circuit representing a non-power in $\pi_1\Gamma$. Attach a 2-cell along λ , obtaining Y . If π_1Y is Baumslag–Solitar-free, then, for every connected subgraph $S \subseteq \Gamma$ with $\lambda(S^1) \not\subseteq S$, the map*

$$\pi_1S \longrightarrow \pi_1Y$$

is injective and its image is quasiconvex.

A choice of basepoint paths changes these subgroups only by conjugacy. Such an S is called a *Magnus subgraph*. Its injectivity follows from the Freiheitssatz. Once Theorem 1.1 gives

ambient hyperbolicity, its quasiconvexity follows from Linton's theorem that Magnus subgroups of hyperbolic one-relator groups are quasiconvex [13, Theorem 4.7]. We prove the graph-to-basis identification explicitly in Lemma 8.1. The same quasiconvexity theorem supplies the quasiconvex edge groups at each inductive step, so the induction needs only to establish hyperbolicity of the smaller vertex group.

1.4 Structure of the proof

Sections 2–7 prove the restricted combination theorem. Section 3 states its hypotheses and converts nonflaring annuli into a bounded bilateral sequence of currents. Section 4 removes atoms and obtains a stationary current with well-defined transfers on a conull set. Sections 5 and 6 construct the finite carriers and exclude periodic support lines. Section 7 finishes the measure contradiction. Finally, Section 8 constructs a decreasing Magnus induction and proves Theorems 1.1 and 1.2. Section 9 derives the virtual free-by-cyclic consequence and records the corresponding ascending-HNN consequence of the companion theorem.

2 Intersections, cyclic commensurators, and rank bounds

We need three preliminary tools. The first controls intersections of quasiconvex subgroups. The second produces an embedded Baumslag–Solitar subgroup from a tree action. The third bounds the total reduced rank of a family of conjugate intersections.

2.1 Quasiconvex intersections

All word metrics come from finite generating sets. A group is *word-hyperbolic* if geodesic triangles in a Cayley graph are uniformly thin: each side lies in a fixed neighborhood of the other two. A subgroup of a hyperbolic group is *quasiconvex* if every geodesic between its elements lies in a fixed neighborhood of the subgroup. We use the Morse lemma and its usual consequences: quasiconvex subgroups are finitely generated and quasi-isometrically embedded, these two properties characterize quasiconvexity, and infinite cyclic subgroups are quasiconvex. In particular, quasiconvexity is transitive between hyperbolic groups, since quasi-isometric embeddings compose. For these background facts, see [2, Theorem III.H.1.7, Corollary III.Γ.3.6(2), and Corollary III.Γ.3.10(1)]. For a subset Q of a hyperbolic group, write ΛQ for its accumulation set in the boundary.

Lemma 2.1 (Quasiconvex intersections). *Let K, L be quasiconvex subgroups of a finitely generated hyperbolic group H . Then $K \cap L$ is quasiconvex and*

$$\Lambda(K \cap L) = \Lambda K \cap \Lambda L.$$

The same assertions hold for any finite family of quasiconvex subgroups.

Proof. First observe that, for fixed subgroups K, L of any finitely generated group and $R \geq 0$, there is a constant D such that

$$N_R(K) \cap N_R(L) \subseteq N_D(K \cap L). \quad (2.1)$$

Indeed, if x belongs to the left side, choose $k \in K$ and $l \in L$ within distance R of x . The element $d = k^{-1}l$ lies in the finite ball of radius $2R$. For each value d that occurs, fix one pair $k_d \in K, l_d \in L$ with $k_d^{-1}l_d = d$. Then

$$z = kk_d^{-1} = ll_d^{-1} \in K \cap L, \quad d(x, z) \leq R + |k_d|.$$

Taking a maximum over the finitely many d proves (2.1).

A geodesic with endpoints in $K \cap L$ lies in fixed neighborhoods of both K and L , so (2.1) proves quasiconvexity of their intersection.

Now let $\xi \in \Lambda K \cap \Lambda L$. Quasiconvexity and limits of geodesic segments show that a ray from the identity to ξ lies in bounded neighborhoods of both subgroups. Applying (2.1) along the ray gives a sequence in $K \cap L$ converging to ξ . Thus $\xi \in \Lambda(K \cap L)$. The reverse inclusion is immediate, and induction gives the finite-family assertions. These facts also appear in [7, Lemmas 2.6 and 2.7]. $\square$

Lemma 2.2 (Finite double-coset lists). *Let K, L be quasiconvex subgroups of a finitely generated hyperbolic group H . There are only finitely many double cosets KgL such that $K \cap gLg^{-1}$ contains an infinite-order element. If H is torsion-free, there are therefore only finitely many such double cosets with nontrivial intersection.*

Proof. Suppose that $c \in K \cap gLg^{-1}$ has infinite order. Its two fixed boundary points belong to both ΛK and $\Lambda(gL)$: use the sequences $c^n \in K$ and $c^n g \in gL$, for positive and negative n . A geodesic line between those points lies in a uniformly bounded neighborhood of each of K and gL . To see the uniformity, a q -quasiconvex set contains finite geodesics between its points within distance q ; passing to limits, and then comparing geodesics with the same ideal endpoints, increases this bound by a constant depending only on the hyperbolicity constant. The quasiconvexity constant of gL equals that of L .

At any point of this line choose nearby vertices $k \in K$ and $gl \in gL$. Their distance is bounded by a constant depending only on the two quasiconvexity constants and the hyperbolicity constant. Hence $k^{-1}gl \in KgL$ lies in a fixed finite ball. Every double coset under consideration meets that ball, proving finiteness. $\square$

The qualification about infinite-order elements is necessary in groups with torsion. For example, the central subgroup C_2 of the hyperbolic group $F_2 \times C_2$ has nontrivial intersection with every conjugate, but there are infinitely many C_2 -double cosets.

For later applications to cosets, note also that

$$\Lambda(gL) = \Lambda(gLg^{-1}).$$

Indeed, right multiplication by the fixed element g^{-1} moves each point of gL a bounded distance.

2.2 Commensuration in a tree action

Two subgroups are *commensurable* if their intersection has finite index in both. For a subgroup $C \leq H$, its commensurator is

$$\text{Comm}_H(C) = \{h \in H : C \text{ and } hCh^{-1} \text{ are commensurable}\}.$$

This is a subgroup. We first deduce the cyclic case directly from the double-coset lemma.

Lemma 2.3. *If C is an infinite cyclic subgroup of a hyperbolic group H , then C has finite index in $\text{Comm}_H(C)$. If H is torsion-free, this commensurator is infinite cyclic.*

Proof. The subgroup C is quasiconvex. By Lemma 2.2, its commensurator meets only finitely many double cosets CgC . For a representative g in the commensurator, CgC is a union of $[C : C \cap g^{-1}Cg]$ left C -cosets, a finite number. Thus the commensurator is virtually cyclic. A torsion-free virtually cyclic group is cyclic, giving the last assertion. $\square$

All tree actions below are actions on simplicial trees without inversions. An element is *elliptic* if it fixes a vertex and *tree-hyperbolic* if it translates a bi-infinite geodesic. We will need an embedded Baumslag–Solitar group, rather than just a power-conjugacy relation. The following normal-form argument provides it.

Lemma 2.4 (Graphs of cyclic groups). *Let Q be a finitely generated noncyclic group acting on a simplicial tree without inversions. If every vertex and edge stabilizer is infinite cyclic, then Q contains a subgroup isomorphic to $BS(m, n)$ for some nonzero integers m, n .*

Proof. Choose a vertex v and a finite symmetric generating set S of Q . The union of all Q -translates of the finite subtree

$$D = \bigcup_{s \in S} [v, sv]$$

is connected: its segments connect the vertices in the orbit Qv according to the Cayley graph of Q . This union is exactly the convex hull of Qv , because its segments join orbit points. Restricting the action to this hull gives a finite quotient, since every edge orbit has a representative in D . The retained vertex and edge stabilizers are unchanged. Bass–Serre theory therefore realizes Q as the fundamental group of a finite connected graph of infinite cyclic groups.

Collapse any non-loop edge whose map to one endpoint group is onto. Such a collapse preserves the fundamental group and leaves a graph of infinite cyclic groups. After finitely many collapses, every remaining non-loop edge has both inclusion indices greater than one.

If a loop remains, the subgraph consisting of that loop and its vertex has fundamental group

$$\langle x, t \mid tx^mt^{-1} = x^n \rangle = BS(m, n), \quad m, n \neq 0.$$

Its fundamental group injects into Q by the graph-of-groups normal form theorem.

If a non-loop edge remains, its two-vertex subgraph likewise injects and has fundamental group

$$\langle x, y \mid x^m = y^n \rangle, \quad |m|, |n| > 1.$$

The element $z = x^m = y^n$ is central and has infinite order. Modulo $\langle z \rangle$, this group is $C_{|m|} * C_{|n|}$, in which the image of xy has infinite order by free-product normal form. Thus the commuting elements z and xy generate an embedded $\mathbb{Z}^2 = BS(1, 1)$.

Finally, if no edge remains, the graph has one vertex and Q is cyclic, contrary to the hypothesis. This exhausts the possibilities. $\square$

Lemma 2.5 (Elliptic commensuration). *Suppose that G acts on a simplicial tree $\mathcal{T}$ without inversions, with torsion-free hyperbolic vertex stabilizers. If an infinite-order elliptic element $a \in G$ has its cyclic subgroup commensurated by a tree-hyperbolic element $g \in G$, then G contains a subgroup isomorphic to $BS(m, n)$ for some nonzero integers m, n .*

Proof. Set $L = \langle a, g \rangle$ and choose a vertex v fixed by a . Every element of L commensurates $\langle a \rangle$. For each $h \in L$, commensurability supplies nonzero integers r, s with $a^r = ha^sh^{-1}$; hence a nonzero power of a fixes hv . A common multiple of finitely many such powers fixes any prescribed finite subset of Lv , and therefore fixes its convex hull pointwise. It follows that every finite subtree of $\mathcal{T}_L = \text{Hull}(Lv)$ is fixed by a nonzero power of a .

Let x be a vertex of $\mathcal{T}_L$, and choose $N > 0$ with $a^N \in L_x$. Every element of L_x commensurates $\langle a^N \rangle$, now a subgroup of the vertex stabilizer G_x . Thus

$$\langle a^N \rangle \leq L_x \leq \text{Comm}_{G_x}(\langle a^N \rangle).$$

Lemma 2.3 makes the last group infinite cyclic, so L_x is infinite cyclic as well. Each edge stabilizer of $\mathcal{T}_L$ contains a nonzero power of a and is a subgroup of a vertex stabilizer in L ; it too is infinite cyclic.

The group L is noncyclic. Otherwise, writing $a = c^r$ and $g = c^s$ in a cyclic group, translation lengths would give $\ell(a) = |r|\ell(c) = 0$ and hence $\ell(g) = 0$, contrary to tree-hyperbolicity of g . Apply Lemma 2.4 to the action of the finitely generated group L on $\mathcal{T}_L$. $\square$

2.3 Strong inertia in free groups

We now formulate the rank estimate that will control noncyclic intersections. For a finitely generated free group K , put

$$\bar{r}(K) = \max\{\text{rk}(K) - 1, 0\}.$$

Strong inertia bounds the sum of reduced ranks of all conjugate intersections; see [12].

Definition 2.6. A finitely generated subgroup E of a finitely generated free group P is *strongly inert* if, for every finitely generated subgroup $S \leq P$,

$$\sum_{SpE \in S \backslash P/E} \bar{r}(E \cap p^{-1}Sp) \leq \bar{r}(S). \quad (2.2)$$

The summands are independent of the chosen double-coset representatives. Only finitely many intersections are nontrivial, by Lemma 2.2.

Corollary 2.7. *Let P be a finitely generated free group and $E \leq P$ a strongly inert subgroup. If $g \in P \setminus E$, then $E \cap gEg^{-1}$ is trivial or cyclic.*

Proof. Apply (2.2) with $S = E$. The identity double coset contributes $\bar{r}(E)$, so all other nonnegative summands vanish. $\square$

3 Annuli and currents for a restricted HNN extension

We now isolate the combination statement used in the Magnus induction. The restrictions on intersections in its hypotheses will provide a uniform bound on the ranks of path stabilizers. In this section we establish that failure of hyperbolicity produces a bounded, nonzero trajectory of free-group currents. The subsequent sections rule out that trajectory and thereby complete the combination argument.

Theorem 3.1 (Restricted combination theorem). *Let H be a torsion-free word-hyperbolic group, let $A, B \leq H$ be finitely generated free quasiconvex subgroups, and let $\phi: A \rightarrow B$ be an isomorphism. Set*

$$G = \langle H, t \mid tat^{-1} = \phi(a) \ (a \in A) \rangle, \quad P = A, \quad I = A \cap B, \quad J = t^{-1}It \leq P.$$

Assume the following two conditions.

- (i) *For $M, N \in \{A, B\}$ and $g \in H$, the intersection $M \cap gNg^{-1}$ is at most cyclic unless $g \in MN$.*
- (ii) *The groups I, J are finitely generated, and each $E \in \{I, J\}$ satisfies*

$$\sum_{SpE \in S \backslash P/E} \bar{r}(E \cap p^{-1}Sp) \leq \bar{r}(S) \quad \text{for every finitely generated } S \leq P. \quad (3.1)$$

If G contains no subgroup isomorphic to $BS(m, n)$ for any nonzero integers m, n , then G is word-hyperbolic.

All the intersection restrictions in this statement are hypotheses. In particular, the assertion concerns a restricted class of HNN extensions. We retain the notation of Theorem 3.1 throughout the combination argument. Write

$$i: I \longrightarrow P, \quad i(c) = c, \quad j: I \longrightarrow P, \quad j(c) = t^{-1}ct,$$

so that $j(I) = J$. Fix finite generating sets for H and a free basis for P . We write $|\cdot|_H$ and $|\cdot|_P$ for the associated based word lengths. If $P = 1$, then $G = H * \mathbb{Z}$ is hyperbolic, so we may assume $P \neq 1$ in what follows.

3.1 Frames and stabilizers of paths

Let $\mathcal{T}$ be the Bass–Serre tree of the HNN extension. The height homomorphism $G \rightarrow \mathbb{Z}$ sends t to 1 and H to 0. Orient the canonical edge e from $t^{-1}H$ to H , and orient every translate equivariantly. Its stabilizer is P . A *frame* for the edge he is the element $h \in G$; changing h to hp , with $p \in P$, changes the coordinates on its stabilizer but not the edge. A path traverses an edge positively if it follows this orientation.

Lemma 3.2 (Neighboring edges and frames). *A reduced path in $\mathcal{T}$ with noncyclic pointwise stabilizer has constant traversal sign. The positive successors of e that can occur in such a path have the form pte , with $pI \in P/I$, and the positive predecessors have the form $pt^{-1}e$, with $pJ \in P/J$. Their common stabilizers with e are, respectively,*

$$pIp^{-1} \quad \text{and} \quad pJp^{-1}.$$

For a positive successor the coordinate transfer is $c \mapsto t^{-1}p^{-1}cpt$; for a positive predecessor it is $c \mapsto tp^{-1}cpt^{-1}$, on the indicated domains.

Proof. Every positively oriented edge leaving H is ate for some $a \in H$; the possible edges are parametrized by $aB \in H/B$. Its common stabilizer with e is $A \cap aBa^{-1}$. If this intersection is noncyclic, assumption (i) gives $a \in AB$. Write $a = pb$ with $p \in P$ and $b \in B$. Since $bt = t\phi^{-1}(b)$, the edge ate is pte . Moreover,

$$pte = qte \iff t^{-1}q^{-1}pt \in A \iff q^{-1}p \in A \cap B = I.$$

The common stabilizer is $P \cap pBp^{-1} = pIp^{-1}$, and conjugating by $(pt)^{-1}$ gives the stated transfer.

A positive predecessor ending at $t^{-1}H$ is $t^{-1}ae$, with $aA \in H/A$. After conjugating its pair stabilizer by t , we obtain $B \cap aAa^{-1}$. Noncyclicity gives $a \in BA$; writing $a = br$ and $p = t^{-1}bt \in P$ puts the predecessor in the form $pt^{-1}e$. Two such edges agree exactly when

$$q^{-1}p \in P \cap t^{-1}Pt = t^{-1}(B \cap A)t = J.$$

Their common stabilizer with e is pJp^{-1} , and conjugation by $(pt^{-1})^{-1}$ is the claimed transfer.

At a reversal of traversal sign, the consecutive edge stabilizers, in the common vertex coordinates, are M and aMa^{-1} for the same $M \in \{A, B\}$. The edges are distinct because the path is reduced, so $a \notin M$. Assumption (i) makes their intersection at most cyclic. A noncyclic stabilizer of the whole path cannot lie in that intersection. Thus there are no reversals. $\square$

We call a consecutive positive pair *aligned* when its bridge lies in the double coset AB in the common vertex coordinates. Reversing the traversal gives the corresponding definition for a consecutive negative pair. Lemma 3.2 says that every path with noncyclic stabilizer is aligned after its traversal sign is fixed.

Lemma 3.3 (Rank bound for a fixed window). *Fix integers $\ell, r \geq 0$. Consider the positively traversed reduced paths with edges indexed by $-\ell, \dots, r$, with edge 0 equal to e , and with noncyclic pointwise stabilizer. Then*

$$\sum_{[W] \in P \setminus \{\text{such paths}\}} \bar{r}(\text{Stab}_G(W)) \leq \bar{r}(P). \quad (3.2)$$

Every stabilizer in this sum is finitely generated and free. In particular, there are at most $\bar{r}(P)$ such P -orbits. Equivalently, the same bound holds for G -orbits if the distinguished edge is allowed to vary.

Proof. Build the fixed window by starting with e and adding one edge at an endpoint at a time. The initial stabilizer is P . Suppose W is an existing representative, $S = \text{Stab}_G(W) \leq P$, and h is a frame at the endpoint edge being extended. In that frame put $K = h^{-1}Sh \leq P$.

For a forward extension whose stabilizer is noncyclic, Lemma 3.2 restricts the next edge to $hpte$, with $pI \in P/I$. The action of S on these choices becomes left multiplication by K . Thus the extension orbits over the orbit of W are indexed by the double cosets KpI , and the extended stabilizer is isomorphic to $K \cap pIp^{-1}$. Assumption (ii) gives

$$\sum_{KpI} \bar{r}(K \cap pIp^{-1}) \leq \bar{r}(K) = \bar{r}(S).$$

For backward extension use J and its separately assumed inequality. The stabilizers remain finitely generated: in the current edge coordinates an extension takes the intersection of two finitely generated subgroups of the finite-rank free group P .

The forgetful map from extended windows to the preceding windows is P -equivariant. Its fiber over the orbit of W is precisely the set of extensions modulo S , which is the double-coset set just considered. Summing the displayed inequality over the existing representatives therefore proves the induction step. Windows with cyclic stabilizer may be discarded, since extension only decreases a stabilizer. This proves (3.2). Each noncyclic term contributes at least one, proving the orbit bound. Finally, $\text{Stab}_G(e) = P$, which identifies the two stated orbit sets. $\square$

The shape (ℓ, r) is fixed in this lemma; no bound is asserted for the union of windows of all lengths. We shall also use the following direct consequence of the same hypotheses.

Lemma 3.4. *Two distinct aligned paths of the same fixed window shape with distinguished edge e and the same traversal sign have at most cyclic common pointwise stabilizer.*

Proof. At their first divergence from the common part, the common stabilizer, in the shared endpoint frame, lies in

$$pEp^{-1} \cap qEq^{-1}, \quad pE \neq qE, \quad E \in \{I, J\}.$$

This intersection is at most cyclic. Indeed, apply (3.1) with $S = E$. The diagonal double coset already contributes $\bar{r}(E)$, so every other term has reduced rank zero. Since $p^{-1}q \notin E$, the displayed intersection is one of these off-diagonal intersections up to conjugacy. If E is cyclic or trivial, the conclusion is immediate as well. $\square$

3.2 The algebraic annular-flaring criterion

We state the external combination theorem with its quantifiers before passing to limits. This is important because the width bound in a sequence of counterexamples will depend on the number of rows.

For a traversal sign $\epsilon \in \{+, -\}$ define the arriving and departing maps into H by

$$f_+(c) = c, \quad f_-(c) = tct^{-1}, \quad l_+(c) = tct^{-1}, \quad l_-(c) = c.$$

An *algebraic annulus with $2m + 1$ rows* consists of signs $\epsilon_s \in \{+, -\}$ and elements $c_s \in P \setminus \{1\}$ for $-m \leq s \leq m$, together with bridges $a_s \in H$ for $-m \leq s < m$, satisfying

$$f_{\epsilon_s}(c_s) = a_s l_{\epsilon_{s+1}}(c_{s+1}) a_s^{-1}. \quad (3.3)$$

Its width is $\max_s |a_s|_H$ and its girth is $|c_0|_P$. It is *essential* if a $+$, $-$ transition has $a_s \notin A$ and a $-$, $+$ transition has $a_s \notin B$. There is no restriction at equal-sign transitions. It is λ -*hyperbolic* if

$$\max\{|c_{-m}|_P, |c_m|_P\} \geq \lambda |c_0|_P.$$

All these lengths are based word lengths.

For completeness, these definitions describe an actual reduced tree path. Write $f_{\epsilon_s}(c) = u^{-1}cu$ and $l_{\epsilon_{s+1}}(c) = v^{-1}cv$, where $u, v \in \{1, t^{-1}\}$. Starting with a frame h_s , choose

$$h_{s+1} = h_s u a_s v^{-1}. \quad (3.4)$$

The two edges meet at $h_s u H = h_{s+1} v H$, and (3.3) gives $h_s c_s h_s^{-1} = h_{s+1} c_{s+1} h_{s+1}^{-1}$. Thus a single nontrivial element fixes the developed path. The four possible frame multipliers are

$$\begin{array}{c|cccc} (\epsilon_s, \epsilon_{s+1}) & (+, +) & (+, -) & (-, +) & (-, -) \\ \hline u a_s v^{-1} & a_s t & a_s & t^{-1} a_s t & t^{-1} a_s. \end{array}$$

At the two reversing transitions the consecutive geometric edges agree exactly when $a_s \in A$ or $a_s \in B$, respectively. Essentiality is therefore exactly absence of backtracking.

Theorem 3.5 (Algebraic annular flaring). *Let H be word-hyperbolic, let $A, B \leq H$ be finitely generated, and let $\phi : A \rightarrow B$ be an isomorphism such that both attaching maps $A \rightarrow H$ are quasi-isometric embeddings. Set $G = \langle H, t \mid tat^{-1} = \phi(a), a \in A \rangle$. Then G is word-hyperbolic if there exist $\lambda > 1$ and an integer $m \geq 1$ such that, for every $\rho > 0$, there is $h(\rho)$ for which every essential algebraic annulus as defined above, with $2m + 1$ rows, width at most ρ , and girth at least $h(\rho)$, is λ -hyperbolic.*

This is the algebraic formulation of the Bestvina–Feighn combination theorem in [1, Theorem 1.2]; we use its explicit annular statement in [8, Definitions 4.1–4.4 and Theorem 4.5]. The latter numbers refer to the arXiv version of that paper. Our $2m + 1$ counts the successive edge groups, including both ends. The criterion applies here because A and B are quasiconvex in the hyperbolic group H .

Lemma 3.6 (Counterannuli at a fixed length). *If G is not hyperbolic, then for every $m \geq 1$ there is a sequence of essential annuli with $2m + 1$ rows whose signs and bridges are fixed throughout the sequence, such that*

$$|c_0|_P \longrightarrow \infty, \quad |c_{-m}|_P, |c_m|_P < 2|c_0|_P.$$

For each fixed $s \in [-m, m]$, the ratios $|c_s|_P / |c_0|_P$ are bounded. The bounds and the fixed bridge tuple may depend on m .

Proof. Negate Theorem 3.5 with $\lambda = 2$. For each m there is a finite width bound ρ_m and counterannuli of arbitrarily large girth with width at most ρ_m . Choose such a sequence with girth tending to infinity. There are finitely many sign tuples and finitely many bridge tuples of this fixed length and width, so a subsequence fixes both tuples.

The maps f_ϵ, l_ϵ are quasi-isometric embeddings. The row equation, with the bridges now fixed, gives constants $L \geq 1$ and $D \geq 0$ such that

$$|c_{s+1}|_P \leq L|c_s|_P + D, \quad |c_s|_P \leq L|c_{s+1}|_P + D$$

at every row of this fixed sequence. Iterating from row 0 bounds all the indicated ratios, since $|c_0|_P \rightarrow \infty$. $\square$

3.3 Currents and induction along an injection

Let F be a nontrivial finite-rank free group with a fixed free basis, and let T_F be its Cayley tree. The space

$$\partial^2 F = (\partial F \times \partial F) \setminus \{(\xi, \xi) : \xi \in \partial F\}$$

parametrizes oriented geodesic lines: (ξ, ζ) is directed from ξ to ζ . For a finite nondegenerate oriented segment γ in T_F , let $\text{Cyl}_F(\gamma)$ be the set of lines containing it with that orientation. These compact-open cylinders form a basis for the topology. Let $\mathcal{L}_F(v)$ be the set of oriented lines passing through the vertex v . It is a finite disjoint union of the cylinders on edges directed out of v , hence is compact open.

A *current* on F is a nonnegative, F -invariant, flip-invariant Radon measure on $\partial^2 F$. Denote its cone by $\text{Curr}(F)$ and give it the topology of convergence on compact-open cylinders. Its norm is

$$\|\mu\|_F = \frac{1}{2}\mu(\mathcal{L}_F(1)). \quad (3.5)$$

This norm is additive on the cone. It vanishes only on the zero current, since translates of $\mathcal{L}_F(1)$ cover $\partial^2 F$. We omit the subscript when the group and basis are clear.

For $w \in F \setminus \{1\}$, the *counting current* η_w is the sum of point masses at the two orientations of the translates of the axis of w , indexed by $F/\langle w \rangle$. Indexing by $\langle w \rangle$ retains the multiplicity when w is a proper power. Directly counting the translated axes through 1 gives

$$\|\eta_w\|_F = |w|_{F, \text{cyc}}, \quad (3.6)$$

where the right side is cyclic word length. More generally, the coordinate on a segment labeled by a reduced word u counts the occurrences of u and u^{-1} in the cyclic word for w , with multiplicity. For a cyclic word z , occurrences are tested at all $|z|$ starting positions in its periodic repetition, including when $|u| > |z|$.

Lemma 3.7 (Compactness). *The norm is continuous on $\text{Curr}(F)$, and each closed norm-bounded subset is compact. In particular, every bounded sequence of currents has a convergent subsequence.*

Proof. Continuity follows because $\mathcal{L}_F(1)$ is compact open. Every compact subset of $\partial^2 F$ is covered by finitely many translates of this set. A uniform norm bound therefore bounds the measures of all compact subsets. There are countably many segment cylinders. Choose a diagonal subsequence on their measures. The cylinder identities are preserved: a cylinder is the finite disjoint union of its extensions by one edge at either end. On each compact-open $\mathcal{L}_F(v)$ these identities define a finite measure, by extension from its clopen cylinders, and the measures agree on overlaps. They

consequently define a Radon measure on line space. Invariance and flip invariance pass to the limit. The continuous norm retains its bound. This proves sequential compactness; the countable cylinder coordinates metrize the topology on a bounded set, so compactness follows. $\square$

We use the current map associated to an injection with its full induction over cosets. An ordinary boundary push alone would generally fail to be invariant under the target group. The construction below agrees with [9, Proposition–Definition 12.1] in the nonabelian case, and also covers rank one.

Lemma 3.8 (Induction of currents). *Let $\theta: Q \rightarrow F$ be an injection of nontrivial finite-rank free groups. It defines a continuous linear map*

$$\theta_*: \text{Curr}(Q) \longrightarrow \text{Curr}(F)$$

which sends η_w to $\eta_{\theta(w)}$ and sends every nonzero current to a nonzero current. For the fixed bases there are constants $0 < c_\theta \leq C_\theta < \infty$ such that

$$c_\theta \|\nu\|_Q \leq \|\theta_*\nu\|_F \leq C_\theta \|\nu\|_Q \quad (\nu \in \text{Curr}(Q)). \quad (3.7)$$

Postcomposing θ with an inner automorphism of F does not change θ_ . Induction is functorial: if $\psi: F \rightarrow F'$ is another injection of nontrivial finite-rank free groups, then*

$$(\psi \circ \theta)_* = \psi_* \circ \theta_*.$$

Proof. Put $E = \theta(Q)$. The subgroup E is finitely generated and hence quasiconvex in F . Thus θ is a quasi-isometric embedding and extends to a homeomorphism $\partial Q \rightarrow \Lambda E \subset \partial F$. Push a current ν by this boundary map to an E -invariant measure $\widehat{\nu}$ on $\partial^2 \Lambda E$, viewed as a measure on $\partial^2 F$, and define

$$\theta_*\nu = \sum_{pE \in F/E} p_*\widehat{\nu}. \quad (3.8)$$

The summand depends only on the coset. It remains to prove that this sum is locally finite and continuous.

Let T_E be the convex hull of ΛE in T_F . Its quotient by E is the finite unbased core graph of E . For a fixed vertex v , there are only finitely many cosets pE for which $v \in pT_E$: indeed such cosets correspond to E -orbits of vertices $p^{-1}v$ in T_E . A coset term in (3.8) that meets a segment cylinder has its hull containing that segment. Hence only finitely many terms contribute to a fixed cylinder. The preimage of that cylinder in $\partial^2 Q$ is compact open, because $\partial^2 \Lambda E$ is closed in $\partial^2 F$ and the boundary map is a homeomorphism onto it. It follows that the sum is Radon and that every cylinder coordinate depends continuously and linearly on ν . The sum is F -invariant and flip-invariant by construction.

The coset decompositions in (3.8) combine the index sets F/E and $E/\theta(\langle w \rangle)$ into $F/\langle \theta(w) \rangle$. This proves the assertion on counting currents, including proper powers. The same coset calculation proves functoriality. Indeed, in $\psi_*\theta_*\nu$ the outer representatives range over $F'/\psi(F)$ and the inner representatives over $\psi(F)/\psi\theta(Q)$. Their products form representatives for $F'/\psi\theta(Q)$, and the boundary maps compose. Thus the two nonnegative sums give exactly $(\psi \circ \theta)_*\nu$.

A nonzero measure has nonzero boundary push, and the summands are nonnegative, so its induction is nonzero. By Lemma 3.7, the norm-one set in $\text{Curr}(Q)$ is compact. The continuous function $\nu \mapsto \|\theta_*\nu\|_F$ on that set is positive, so its minimum and maximum give (3.7). Finally, conjugating E and the boundary map simply reindexes the coset sum; the resulting F -invariant current is unchanged. $\square$

No injectivity of the linear map θ_* is asserted or needed. Nonzero preservation and the two norm inequalities are the properties that will be used.

3.4 From based words to currents

The girth of an annulus uses based length, whereas the norm of a counting current uses cyclic length. A conjugating tail can make these lengths very different. We bridge that difference by capping the word in the intersection subgroup before applying either of its two embeddings.

For reduced words u, w over a fixed basis, let $N_u(w)$ count the occurrences of u as a contiguous subword of the linear word w . Occurrences crossing an end are not counted. Write $N_u^{\text{sym}}(w) = N_u(w) + N_{u^{-1}}(w)$.

Lemma 3.9 (Capping linear words). *Fix an injection $\theta: Q \rightarrow F$ of nontrivial finite-rank free groups with fixed bases. For each nonempty reduced word W in Q , there is a suffix b of length at most one such that Wb is cyclically reduced. For every fixed nonempty reduced word u in F there is a constant $D_{\theta,u}$, independent of W , such that*

$$\left| \eta_{\theta(Wb)}(\text{Cyl}_F([1, u])) - N_u^{\text{sym}}(\text{red}(\theta(W))) \right| \leq D_{\theta,u}. \quad (3.9)$$

There is also a constant D_θ independent of W such that

$$||\eta_{\theta(Wb)}||_F - |\theta(W)|_F \leq D_\theta. \quad (3.10)$$

Here red denotes the unique reduced representative. The same cap b may be used simultaneously for any finite collection of fixed injections out of Q .

Proof. If Q is cyclic, every nonempty reduced word is already cyclically reduced, so take b empty. Otherwise there are at least four oriented basis letters. Choose a letter b different from the inverse of the last letter of W and from the inverse of its first letter. Then Wb is freely and cyclically reduced. This choice depends only on Q and W , not on θ .

The image in T_F of a geodesic in T_Q , with the images of successive vertices joined by geodesic segments, is a uniform quasigeodesic. Its constants depend only on θ and the bases. The axis of Wb in T_Q passes through 1. Its image is therefore a uniform quasigeodesic line passing through 1, with endpoints the fixed points of $\theta(Wb)$. The Morse lemma gives a constant D , independent of W , bounding the distance from 1 to the axis of $\theta(Wb)$ in T_F . Accordingly, if

$$\text{red}(\theta(Wb)) = vzv^{-1}$$

with z cyclically reduced and the concatenation reduced, then $|v|_F \leq D$.

Multiplication by $\theta(b)$ changes only a bounded suffix of $\text{red}(\theta(W))$: at most $\max\{|\theta(x)|_F : x \text{ a basis letter of } Q\}$ letters can be canceled, and at most that many can be added. Removing v and v^{-1} then deletes at most $2D$ additional letters. For a fixed word u , changing or deleting a bounded number of letters at the ends changes its linear occurrence count by a bounded amount. Passing from a linear word z to its cyclic occurrence count changes the count by at most $|u| - 1$ at each chosen orientation, even when $|z| < |u|$. These observations prove (3.9), including the inverse occurrences. The same bounded additions and deletions show that $|z|_F = |\theta(W)|_F + O_\theta(1)$; now use (3.6) to obtain (3.10). All constants can be chosen for each member of a finite collection of injections while retaining the same cap. $\square$

Lemma 3.10 (A current limit for each finite window). *Suppose G is not hyperbolic, and fix $m \geq 1$. There are currents $\mu_s \in \text{Curr}(P)$ for $-m \leq s \leq m$ with*

$$\|\mu_0\| = 1, \quad \|\mu_{-m}\|, \|\mu_m\| \leq 2, \quad (3.11)$$

and a fixed essential sign and bridge tuple such that each consecutive pair is induced by a current on its intersection subgroup. More precisely, for

$$K_s = f_{\epsilon_s}(P) \cap a_s l_{\epsilon_{s+1}}(P) a_s^{-1}$$

there is $\nu_s \in \text{Curr}(K_s)$ satisfying

$$(\theta_{s,0})_* \nu_s = \mu_s, \quad (\theta_{s,1})_* \nu_s = \mu_{s+1}, \quad (3.12)$$

where

$$\theta_{s,0} = f_{\epsilon_s}^{-1}, \quad \theta_{s,1} = l_{\epsilon_{s+1}}^{-1} \circ (c \mapsto a_s^{-1} c a_s).$$

All the groups K_s are nontrivial finitely generated free groups.

Proof. Take the sequence from Lemma 3.6, keeping m , all signs, and all actual bridges fixed. Write its elements as $c_s^{(k)}$ and put $L_k = |c_0^{(k)}|_P \rightarrow \infty$. Lemma 2.1 shows that K_s is finitely generated. It is free because it lies in the free group $f_{\epsilon_s}(P)$, and is nontrivial because it contains $f_{\epsilon_s}(c_s^{(k)})$. Fix a basis for each K_s .

Represent $f_{\epsilon_s}(c_s^{(k)})$ by a reduced word $W_s^{(k)}$ in the chosen basis of K_s . Apply Lemma 3.9 to choose a cap $b_s^{(k)}$, and form

$$\nu_s^{(k)} = L_k^{-1} \eta_{W_s^{(k)} b_s^{(k)}}.$$

The injection $\theta_{s,0}$ is quasi-isometric between free groups. Consequently $|W_s^{(k)}|_{K_s}$ is at most a fixed multiple of $|c_s^{(k)}|_P$ plus a fixed constant. The ratio bound from Lemma 3.6 shows that $\|\nu_s^{(k)}\|$ is bounded. There are only finitely many s , so after passing to a subsequence all $\nu_s^{(k)}$ converge, to currents ν_s .

Apply Lemma 3.9 to both injections from K_s . The row equation gives

$$\theta_{s,0}(W_s^{(k)}) = c_s^{(k)}, \quad \theta_{s,1}(W_s^{(k)}) = c_{s+1}^{(k)}.$$

For every fixed word cylinder, the two induced normalized currents therefore differ from the corresponding normalized symmetrized linear row frequencies by $O(1)/L_k$. At an interior row, the two neighboring induced limits agree, since both are limits of the same linear frequencies in $c_s^{(k)}$. Denote the resulting current by μ_s ; at the end rows use their single adjacent transitions. Continuity from Lemma 3.8 gives (3.12).

The length conclusion (3.10) gives

$$\|\mu_s\| = \lim_k \frac{|c_s^{(k)}|_P}{L_k}$$

along this subsequence. The middle ratio is 1 and the two end ratios are less than 2, proving (3.11). The errors used here may depend on m , the actual bridges, the bases of K_s , and the test cylinder. They disappear because m and all those data were fixed before $L_k \rightarrow \infty$. $\square$

3.5 Finite transition types and a bilateral limit

The preceding construction produces one finite current trajectory for each m . To pass to unbounded window lengths, we first replace the actual bridges by a finite list of double-coset representatives. This replacement is made after the current limits, so it does not require any bound on the original bridges that is uniform in m .

For each ordered sign pair (ϵ, δ) , choose representatives d for the double cosets

$$f_\epsilon(P) \backslash H / l_\delta(P)$$

with nontrivial intersection

$$K_{\epsilon, \delta, d} = f_\epsilon(P) \cap dl_\delta(P)d^{-1}.$$

There are finitely many, by Lemma 2.2 and torsion-freeness of H . Use $d = 1$ for the identity double coset when its intersection is nontrivial. For reversing sign pairs discard that double coset, since it gives precisely the forbidden pinches. Call the resulting finite set of triples $\tau = (\epsilon, \delta, d)$ the *transition types*, and write

$$K_\tau = K_{\epsilon, \delta, d}, \quad \theta_{\tau, 0} = f_\epsilon^{-1}, \quad \theta_{\tau, 1} = l_\delta^{-1} \circ (c \mapsto d^{-1}cd). \quad (3.13)$$

Fix a free basis for every K_τ .

Lemma 3.11 (Uniformity of transition types). *Every consecutive pair of currents from Lemma 3.10 can be expressed as*

$$\mu_s = (\theta_{\tau_s, 0})_* \nu_s, \quad \mu_{s+1} = (\theta_{\tau_{s+1}, 1})_* \nu_s$$

for a current on one of the fixed groups K_{τ_s} , with type signs $(\epsilon_s, \epsilon_{s+1})$. There are constants $C \geq 2$ and $D \geq 1$, independent of m , such that

$$C^{-1} \|\mu_s\| \leq \|\mu_{s+1}\| \leq C \|\mu_s\|, \quad \|\nu_s\| \leq D \|\mu_s\|. \quad (3.14)$$

If K_τ is noncyclic, the signs are equal and $d = 1$. For the positive aligned type the two maps are i, j , and for the negative aligned type they are j, i .

Proof. Suppress the row index and write $f = f_\epsilon$, $l = l_\delta$. Factor an actual bridge as $a = f(p)dl(q)$ with $p, q \in P$. Conjugation by $f(p)$ identifies $K_d = f(P) \cap dl(P)d^{-1}$ with $K_a = f(P) \cap al(P)a^{-1}$. Under this identification, the maps from K_a to the two edge coordinates become

$$k \mapsto p\theta_{\tau, 0}(k)p^{-1}, \quad k \mapsto q^{-1}\theta_{\tau, 1}(k)q.$$

By Lemma 3.8, these inner automorphisms leave the induced currents unchanged. Transporting the transition current along the group isomorphism therefore gives the desired representation. Essentiality is unchanged: for a reversal, membership of the bridge in the common side subgroup depends only on its double coset.

Each of the finitely many maps in (3.13) has upper and positive lower norm bounds from Lemma 3.8. Comparing the two bounds for a type and then taking maxima over the finite type set proves (3.14).

If K_τ is noncyclic, assumption (i) puts d in the identity double coset. At a reversal that coset was discarded, so the signs must be equal. For equal signs the intersection is I , and direct substitution in (3.13) gives i, j for $(+, +, 1)$ and j, i for $(-, -, 1)$. $\square$

Proposition 3.12 (The current trajectory forced by nonflaring). *Under the hypotheses of Theorem 3.1, if G is not hyperbolic, there exist currents $\mu_s \in \text{Curr}(P)$ for $s \in \mathbb{Z}$, signs $\epsilon_s \in \{+, -\}$, and transition types τ_s with signs $(\epsilon_s, \epsilon_{s+1})$, such that*

(i) $\|\mu_0\| = 1$, every μ_s is nonzero, and $\sup_{s \in \mathbb{Z}} \|\mu_s\| < \infty$;

(ii) for every s there is a current $\nu_s \in \text{Curr}(K_{\tau_s})$ with

$$\mu_s = (\theta_{\tau_s, 0})_* \nu_s, \quad \mu_{s+1} = (\theta_{\tau_s, 1})_* \nu_s;$$

(iii) all types are essential: any reversing type uses a double coset outside the common side subgroup.

The constants in (3.14) apply to this trajectory. In particular, $\|\mu_s\| \geq C^{-|s|}$ for all s .

Proof. For each m take one finite current trajectory from Lemma 3.10, express its transitions using Lemma 3.11, and put

$$D_m = \max_{-m \leq s \leq m} \|\mu_s\|.$$

The constants C, D in (3.14) are now independent of m .

Suppose first that D_m is bounded on a subsequence $m \rightarrow \infty$. Keep its original centers. On each fixed finite window the row currents have bounded norm, as do the transition currents by (3.14). The sign and type sets are finite. Successively extract subsequences on the windows $[-n, n]$, fixing the signs and types there and taking limits of the finitely many row and transition currents. A diagonal subsequence gives currents and compatible signs and types for all $s \in \mathbb{Z}$. Continuity of the induced maps preserves the two current equalities. Norm continuity preserves $\|\mu_0\| = 1$ and the uniform upper bound.

In the remaining case choose a subsequence with $D_m \rightarrow \infty$, and choose an index j_m where the maximum is attained. Divide all row and transition currents in that finite trajectory by D_m , and reindex so that j_m becomes 0. The new row norms are at most 1, and the central norm is 1. Both original end norms were at most 2. Iterating the adjacent norm comparison therefore gives

$$D_m \leq 2C^{j_m+m}, \quad D_m \leq 2C^{m-j_m}.$$

Thus

$$\min\{j_m + m, m - j_m\} \geq \log_C(D_m/2) \rightarrow \infty. \quad (3.15)$$

Every fixed bilateral window is eventually present in the recentered trajectory. The same diagonal extraction applies; the transition norms remain uniformly bounded after normalization. It produces a bilateral trajectory with central norm 1 and all row norms at most 1.

In either case, the finite-type equalities and their norm comparisons survive the limit. Applying the lower comparison repeatedly from row 0 gives $\|\mu_s\| \geq C^{-|s|}$, so none of the row currents vanishes. Compatibility of the two signs of a transition with its neighboring rows survives because these discrete data were fixed on each extracted finite window. The forbidden reversing types were excluded from the type set at the outset. This proves all three conclusions. $\square$

There are two distinct limits in this proof. At each fixed window length, first send the original annular girth to infinity, with all actual bridges fixed. Only after those errors have disappeared do we replace bridges by finitely many types and let the window length grow. The quantitative estimate (3.15) ensures that recentering cannot lose either half of the trajectory, while continuity of the norm preserves nonzero mass. We have therefore reduced Theorem 3.1 to excluding the bounded current trajectory of Proposition 3.12 under BS-freeness.

4 A stationary current and its line trajectories

We retain the restricted HNN splitting and notation of Theorem 3.1. Thus

$$G = \langle H, t \mid tAt^{-1} = B \rangle, \quad P = A, \quad I = A \cap B,$$

and $i, j : I \longrightarrow P$ are inclusion and conjugation by t^{-1} , with respective images I, J . In this section G is assumed to contain no Baumslag–Solitar subgroup. Our first goal is to replace the bounded sequence of currents supplied by Proposition 3.12 with one nonzero current satisfying

$$i_*\nu = j_*\nu = \mu.$$

We will then interpret this equality as a measure-preserving transfer of almost every line along a unique path in the Bass–Serre tree $\mathcal{T}$.

4.1 Periodic axes and a finite set of exceptional roots

A *root of an axis* in the Cayley tree of P means a generator of its stabilizer in P . Such a generator is not a proper power; it need not belong to a free basis. We consider roots up to conjugacy and inversion, or equivalently consider the corresponding conjugacy classes of maximal cyclic subgroups.

Lemma 4.1. *Every atom of a free-group current is a periodic axis. If the mass of one oriented axis is $b > 0$ and its root is r , then its translate orbit, together with the reversed orientations, contributes $b|r|_{\text{cyc}}$ to the current norm.*

Proof. Translate successive vertices of an atomic line to the identity. All resulting lines have the same positive mass and pass through the identity. That set of lines is compact and has finite measure, so two of the translates coincide. A nontrivial element of the free group therefore preserves the original line. An element reversing a line would fix a vertex or invert an edge, which is impossible for the free action on the Cayley tree. The line stabilizer consequently acts by translations and is infinite cyclic.

For its root r , the number of distinct translates of one orientation through the identity is $|r|_{\text{cyc}}$. This is the number of vertices in the quotient axis by $\langle r \rangle$. Flip invariance gives the same mass to the other orientation. The definition of the norm as half the mass of the lines through the identity gives the claimed contribution. $\square$

Recall that a transition type $\tau = (\epsilon, \delta, d)$ consists of an intersection group K_τ and two injections

$$\theta_{\tau,0}, \theta_{\tau,1} : K_\tau \longrightarrow P.$$

There are finitely many types in Proposition 3.12. Their traversal signs and double-coset representatives are part of the type. A noncyclic type must be aligned: its two signs agree, its representative is 1, and its two maps are i, j , in one of the two orders.

Lemma 4.2. *There is a finite set $\mathcal{D}$ of root classes with the following properties.*

- (i) *The root of either image of a nontrivial element of a cyclic transition group belongs to $\mathcal{D}$.*
- (ii) *If a periodic axis is carried by $p\Lambda E$ and $q\Lambda E$ for distinct cosets $pE \neq qE$, where $E = I$ or $E = J$, its root belongs to $\mathcal{D}$.*

- (iii) *If an axis root in a noncyclic transition group maps to a proper power under either transition injection, the root of that image belongs to $\mathcal{D}$.*

Here “carried” means that both endpoints lie in the indicated translate of the limit set.

Proof. For each nontrivial cyclic transition group, include the roots of the images of one generator under both injections. This makes a finite set and proves (i).

For $E = I, J$, the strong inertia inequality with test subgroup E implies that every off-diagonal intersection $E \cap xEx^{-1}$, $x \notin E$, is at most cyclic: the diagonal term already contributes $\bar{r}(E)$. By Lemma 2.2, only finitely many double cosets ExE have nontrivial intersection. For each such off-diagonal double coset, include the root in P of a generator of its intersection.

Suppose that an axis is carried by $p\Lambda E$ and $q\Lambda E$, with $pE \neq qE$. Translate its endpoints by p^{-1} and use Lemma 2.1 to obtain

$$\Lambda E \cap \Lambda((p^{-1}q)E(p^{-1}q)^{-1}) = \Lambda(E \cap (p^{-1}q)E(p^{-1}q)^{-1}).$$

The intersection subgroup is nontrivial cyclic. If $p^{-1}q = e_1xe_2$, with $e_1, e_2 \in E$ and x a chosen double-coset representative, it is conjugate by e_1 to $E \cap xEx^{-1}$. Its axis root is therefore one of the included classes. This proves (ii).

Finally, suppose an axis root u has image r^k , where r is a root in P and $|k| > 1$. Write E for the image subgroup, which in a noncyclic transition is I or J . We have $r \notin E$, since otherwise the inverse image of r would exhibit u as a proper power. The distinct cosets E, rE both carry the axis of r . Assertion (ii) proves (iii). $\square$

We next record exactly how a repetition in this finite set produces a group-theoretic obstruction. The observation applies even when the traversal signs change between the two repeated positions.

Lemma 4.3. *Let (h_se) be a reduced edge path in $\mathcal{T}$, indexed by consecutive integers, and let $r_s \in P$ be nontrivial. Suppose the cyclic subgroups*

$$h_s\langle r_s \rangle h_s^{-1}$$

at consecutive positions are commensurable. If $i < j$ have the same traversal sign and $\langle r_i \rangle, \langle r_j \rangle$ are conjugate in P , then G contains a Baumslag–Solitar subgroup.

Proof. Choose $p \in P$ with $p\langle r_i \rangle p^{-1} = \langle r_j \rangle$, and put $g = h_j p h_i^{-1}$. Since p fixes e , this element takes the edge $h_i e$ to $h_j e$. It takes the cyclic subgroup at position i to the subgroup at position j , which is commensurable with the first one.

Let σ be the segment from the midpoint of $h_i e$ to the midpoint of $h_j e$, in the order of the given path. At each join of two consecutive translates $g^k \sigma$, the incoming and outgoing segments use opposite halves of the same edge. Indeed, g preserves the canonical edge orientation, and the two positions have the same traversal sign. Thus these translates concatenate without backtracking to a bi-infinite geodesic. The element g translates that geodesic a positive distance and is tree-hyperbolic.

The cyclic subgroup at position i is infinite and elliptic. Lemma 2.5 applies to it and to g , and gives the asserted embedded Baumslag–Solitar subgroup. $\square$

Proposition 4.4. *Let $(\mu_s)_{s \in \mathbb{Z}}$ be a bounded sequence of nonzero P -currents with transition currents and compatible essential types as in Proposition 3.12. If G is Baumslag–Solitar free, every μ_s is atomless.*

Proof. Suppose a row has an atom, and start indexing there. At each forward transition, the coset formula for the induced current has a positive contribution to this atom from an atom of the transition current. Choose one such atom and take its image in the next row. Local finiteness of the coset sum and injectivity of the boundary maps in Lemma 3.8 justify this choice. Lemma 4.1 makes every chosen line a periodic axis. Let r_s be its actual root in the coordinates of the s th edge.

We first develop these matches into an actual reduced tree path, so that the repetition criterion of Lemma 4.3 will apply. For a transition with representative d , use the arriving and departing embeddings $f, l : P \rightarrow H$ from the annulus description. A generator $k \in K_\tau$ of the stabilizer of the chosen upstairs axis has images of the form

$$\theta_{\tau,0}(k) = p^{-1}r_s^a p, \quad \theta_{\tau,1}(k) = q^{-1}r_{s+1}^b q, \quad a, b \in \mathbb{Z} \setminus \{0\}.$$

The bridge

$$f(p)dl(q)^{-1}$$

therefore matches r_s^a with r_{s+1}^b . It belongs to the same double coset as d , so it preserves essentiality at a reversal. Develop these bridges using the frame recursion (3.4). They give a reduced path with the prescribed signs, and the framed cyclic subgroups are commensurable at every step. This construction works for arbitrary nonzero exponents, including negative ones. It therefore suffices to find two positions with the same root class and traversal sign.

Call a transition exceptional for this chosen trajectory if its group is cyclic, if either chosen image line is carried by more than one coset of its image subgroup, or if an upstairs root has a proper-power image. By Lemma 4.2, at least one of the adjacent roots at every exceptional transition belongs to $\mathcal{D}$. If infinitely many transitions are exceptional, there are infinitely many such root occurrences. Some root class and some traversal sign occur together at two different positions.

If there are only finitely many exceptional transitions, discard an initial segment. On the remaining tail there is exactly one contributing coset on each side of every push formula. Hence both image atoms have exactly the mass of their common upstairs atom. The chosen atoms have one fixed positive mass on this tail. The norm bound and Lemma 4.1 give a uniform bound on $|r_s|_{\text{cyc}}$. There are finitely many conjugacy classes of roots of bounded cyclic length, so once again a root class repeats at two positions with the same traversal sign.

In either case, Lemma 4.3 gives a Baumslag–Solitar subgroup, a contradiction. $\square$

4.2 A stationary current

We now have an atomless bounded trajectory. The next argument turns it into a stationary one; the case in which the total mass is summable must be treated separately from averaging.

Proposition 4.5. *Under the assumptions of Proposition 4.4, there exist a nonzero current ν on I and a nonzero atomless current μ on P such that*

$$i_*\nu = j_*\nu = \mu. \tag{4.1}$$

In particular, if the restricted HNN extension is not hyperbolic, Proposition 3.12 supplies such a stationary current.

Proof. A nonzero current on an infinite cyclic group induces an atomic current. Thus no transition in the atomless trajectory is cyclic. Every transition is consequently aligned, all traversal signs are

the same, and its maps are i, j or j, i according to that sign. Reverse the indexing if necessary. We then have currents ν_s on the same group I satisfying

$$i_*\nu_s = \mu_s, \quad j_*\nu_s = \mu_{s+1}.$$

In particular I is nontrivial. Fix a free basis of I for its current norm, and define

$$S_N = \sum_{s=-N}^N \nu_s, \quad a_N = \|S_N\| = \sum_{s=-N}^N \|\nu_s\|.$$

The last equality uses positivity. Telescoping gives

$$j_*S_N - i_*S_N = \mu_{N+1} - \mu_{-N}. \quad (4.2)$$

First suppose $a_N \rightarrow \infty$. By compactness of the norm-one current space, pass to a subsequence on which S_N/a_N converges to a current ν of norm one. A uniform norm bound controls the mass of any fixed compact cylinder: such a cylinder is contained in finitely many translates of the set of lines through the identity. The right side of (4.2), divided by a_N , therefore tends to zero on every compact cylinder. Continuity of the induced-current maps yields $i_*\nu = j_*\nu$. Their positive lower norm bounds show that this common current μ is nonzero.

Now suppose (a_N) is bounded. The positive bilateral sum

$$\nu = \sum_{s \in \mathbb{Z}} \nu_s$$

has finite positive norm and defines a Radon measure: every compact line set is covered by finitely many translates of the identity line set. Monotone convergence permits induction through this sum. Reindexing the resulting positive series gives

$$i_*\nu = \sum_{s \in \mathbb{Z}} \mu_s = \sum_{s \in \mathbb{Z}} \mu_{s+1} = j_*\nu.$$

These currents have finite norm by the upper norm bound for induction, and are nonzero. Equivalently, summability and norm comparison make the endpoint norms in (4.2) tend to zero.

In either case we must still prove atomlessness: a weak limit of atomless currents can have atoms. Use the constant bilateral trajectory with row μ and transition current ν . Its transitions are aligned and positive, it is bounded, and Proposition 4.4 applies. Thus μ is atomless. Every atom of ν would contribute positive atomic mass through an injective boundary map in the nonnegative coset formula. Thus ν is atomless as well. $\square$

4.3 Transfers on a conull set of lines

Fix the currents in (4.1), and let $\Omega = \text{supp } \mu \subset \partial^2 P$ be the closed support in line space. A boundary point is *rational* if a nontrivial element of the free group fixes it. Equivalently, its ray labels are eventually periodic. We first remove all lines with a rational endpoint.

Lemma 4.6. *For an atomless free-group current, almost every line has two irrational endpoints. An injection between finitely generated free groups preserves and reflects rationality of boundary points.*

Proof. Fix a rational endpoint ξ , and orient an axis toward it. Every oriented line with positive endpoint ξ , other than this axis, merges with the axis at a first vertex. Fix a possible merge vertex v , and let D_v be this set of lines. If r is the axis root directed toward ξ , the sets $r^{-n}D_v$, $n \geq 0$, are pairwise disjoint. They all pass through v , because their merge vertices are $r^{-n}v$. They have equal measure by invariance, so finiteness of the measure on the compact line set through v gives $\mu(D_v) = 0$. There are countably many merge vertices and rational endpoints. Atomlessness removes the exceptional axes, and flip invariance treats negative endpoints.

An injection preserves rationality by equivariance of its boundary map. To prove reflection, identify its image with a finitely generated subgroup E of a free group. Suppose $\xi \in \Lambda E$ is fixed by a nontrivial element r of the ambient free group. Orient r toward ξ , and choose an axis vertex x . The vertices $r^n x$ on the resulting axis ray are uniformly close to E , by quasiconvexity and the fact that $\xi \in \Lambda E$. Choose $e_n \in E$ at uniformly bounded distance from $r^n x$. Among the finitely many possibilities for $(r^n x)^{-1}e_n$, two agree, say at $n > m$. It follows that

$$e_n e_m^{-1} = r^{n-m} \in E.$$

The inverse image of this nontrivial element fixes the original boundary point, proving reflection. $\square$

We write $\partial^2 \Lambda E$ for the ordered pairs of distinct points of ΛE . If $E = I$ or J , an irrational endpoint belongs to at most one translate $p\Lambda E$, with translates indexed by left cosets pE . Indeed, for two distinct cosets the intersection of the corresponding conjugate subgroups is at most cyclic, and Lemma 2.1 identifies the intersection of their limit sets. Every point of that intersection would be rational.

Proposition 4.7. *There is a translation- and flip-invariant conull Borel set $\mathcal{U} \subset \Omega$ with the following properties.*

- (i) *Every line of $\mathcal{U}$ has two irrational endpoints and is carried by a unique I coset and a unique J coset.*
- (ii) *On lines carried by $p\Lambda I$, translate by p^{-1} and apply the boundary isomorphism from I to J . This forward transfer, and the analogous backward transfer from $q\Lambda J$, preserve μ on their domains.*
- (iii) *Every defined transfer, its inverse, and every output frame translation take lines of $\mathcal{U}$ to lines of $\mathcal{U}$. Consequently all successive forward and backward transfers exist on $\mathcal{U}$.*

Measure preservation in (ii) holds on the full Borel domain of lines with irrational endpoints, not only after restriction to $\mathcal{U}$.

Proof. Let $X \subset \partial^2 P$ consist of lines with two irrational endpoints. It is a conull Borel set by Lemma 4.6. The coset formulas for $i_*\nu$ and $j_*\nu$ show that almost every line of X is carried by some I coset and some J coset. The preceding uniqueness observation proves (i) wherever the carriers exist.

Let $\vartheta : \partial I \rightarrow \partial J$ denote the boundary isomorphism induced by j , with I identified with its included image under i . On the Borel domain

$$D_p = X \cap \partial^2(p\Lambda I)$$

define $F_p(L) = \vartheta(p^{-1}L)$. Rationality preservation and reflection give $F_p(D_p) = X \cap \partial^2 \Lambda J$, a Borel set. For every Borel $D \subset D_p$, the two push formulas each have just one contributing coset on the indicated sets, and give

$$\mu(D) = \nu(p^{-1}D) = \mu(F_p D). \quad (4.3)$$

The middle term is expressed in the boundary coordinates of I . These maps are homeomorphisms between their domains and images in the corresponding limit-set spaces, so the images in this formula are Borel. The same argument applies to the backward branches. Inverse branches preserve measure as well. Output translations and flip preserve it by the definition of a current. Different representatives of one coset merely insert an output translation.

For completeness, we now choose a single conull set on which all these operations can be iterated. Start in X with the Borel null set

$$\begin{aligned} B = & (X \setminus \Omega) \\ & \cup \{L \in X : L \text{ has no } I\text{-coset carrier}\} \\ & \cup \{L \in X : L \text{ has no } J\text{-coset carrier}\}. \end{aligned}$$

The complement of the support is null because line space has a countable base. Saturate B under the partial forward and backward branches, their inverses, translations, and flip. There are countably many such maps and countably many finite compositions. Equation (4.3) shows that each composition takes null sets to null sets on its domain. Their saturation is therefore Borel and null. Its complement in X is the required $\mathcal{U}$. Including $X \setminus \Omega$ among the initial discarded sets ensures that all these iterates remain in the support. $\square$

The set $\mathcal{U}$ is dense in Ω , but it has not been asserted to be closed. Until a separate support argument is given, statements about successive transfers apply to $\mathcal{U}$.

4.4 Actual paths and finite windows

We next describe precisely which tree path is followed by a line in $\mathcal{U}$. This also connects the current to the finite-rank window stabilizers that will be used in the next section.

Proposition 4.8. *Every line $L \in \mathcal{U}$ determines an actual aligned bilateral path of positively traversed edges*

$$\cdots, E_{-1}, E_0 = e, E_1, \cdots$$

in $\mathcal{T}$, with transferred lines in its successive edge frames. The following properties hold.

- (i) *The path is independent of coset representatives and of compatible changes of frames and line coordinates.*
- (ii) *For every finite window containing E_0 , its stabilizer $K \leq P$ is finitely generated and noncyclic, and the endpoints of L lie in ΛK .*
- (iii) *Two distinct actual aligned windows of the same shape about e have at most cyclic common stabilizer. In particular, an irrational point can belong to the limit sets of at most one such window stabilizer.*
- (iv) *If an aligned one-sided window joins e to he , its stabilizer is $S = P \cap hPh^{-1}$. The boundary map of*

$$S \longrightarrow h^{-1}Sh, \quad x \longmapsto h^{-1}xh,$$

implements its line transfer. On the irrational-endpoint domain it preserves μ ; it takes its domain in $\mathcal{U}$ into $\mathcal{U}$.

In (iii), a shape specifies the numbers of edges before and after the central edge, all traversed positively.

Proof. In a frame h for the current edge, let pI be the unique forward carrier coset for its line coordinates. Choose the next edge to be $hpte$, and use the forward transfer for its line coordinates. For the backward extension, the unique coset pJ gives $hpt^{-1}e$. Proposition 4.7 permits indefinite iteration in both directions. The frame description in Lemma 3.2 shows that these are aligned positive paths when ordered from left to right. In particular they are reduced.

Changing p to pi_0 in a forward step, with $i_0 \in I$, does not change the next edge, since

$$pi_0t = ptj(i_0).$$

It changes its frame on the right by $j(i_0) \in P$ and its line coordinates by the inverse translation. Subsequent carrier cosets change by the same coordinate translation, leaving subsequent actual edges unchanged. The backward calculation uses J and the inverse isomorphism. Arbitrary changes of an output frame by P have the same effect. For example, after a forward step to frame hpt , the new line lies in $\partial^2\Lambda J$, so its backward choice is the identity J coset. The backward edge is then $hptt^{-1}e = hpe = he$, with the original line translated into frame hp . This proves (i), including compatibility of the forward and backward constructions.

To prove (ii), build a finite window one end at a time. Initially its stabilizer is P . In the frame at the end being extended, let the current stabilizer be K' . A forward extension intersects it with pIp^{-1} , and a backward extension intersects it with pJp^{-1} . Both subgroups are finitely generated subgroups of a free group. Their intersection is finitely generated, and Lemma 2.1 says that its limit set is the intersection of their limit sets. The current endpoints belong to both, by induction and by the carrier choice. They therefore belong to the new stabilizer limit set. Conjugation by the change of frame transfers this assertion to the next coordinates; its boundary map restricts compatibly to every smaller subgroup. The same intersection computation, without the endpoint assertion, proves finite generation for every finite aligned window. Induction proves the endpoint assertion in the central coordinates. A trivial or cyclic subgroup cannot carry an irrational endpoint, so the stabilizer is noncyclic.

For (iii), Lemma 3.4 gives the cyclic common stabilizer at the first divergence in either direction. The window stabilizers are finitely generated, so their limit sets intersect in the limit set of their intersection. This proves the assertion about irrational endpoints as well. The same proof compares restrictions of longer paths to any common finite shape.

An element fixing the two endpoint edges of a tree segment fixes every edge between them. Thus the stabilizer in (iv) is precisely $P \cap hPh^{-1}$. Successive coordinate changes along the path compose to conjugation by h^{-1} . Their boundary maps agree with the boundary map on this subgroup, giving the asserted formula. If an irrational line has its endpoints in ΛS , every successive carrier needed by this prescribed path exists. Its uniqueness forces the successive branch choices to be exactly those of the path. The measure-preserving branch identities in Proposition 4.7 therefore compose to measure preservation for this partial conjugation. The invariance of $\mathcal{U}$ gives the last assertion. $\square$

Lemma 4.9. *Fix a finite positive window shape about e and a finite ball in the Cayley tree of P . There are only finitely many actual windows of that shape whose stabilizer is noncyclic and whose stabilizer limit set has convex hull meeting that ball.*

Proof. The window bound, Lemma 3.3, gives finitely many P -orbits of such windows. Choose representatives $W_1, \dots, W_k$ with stabilizers $K_1, \dots, K_k$. For each i , the quotient of $\text{Hull}(\Lambda K_i)$ by K_i is a finite core graph, since K_i is finitely generated. Choose finitely many vertices $v_{i,j}$ representing its vertex orbits.

If the hull associated to pW_i meets the fixed ball, enlarge that ball by one edge if necessary and choose a vertex x in the intersection. Then $p^{-1}x = kv_{i,j}$ for some $k \in K_i$ and some j . Consequently

$$pW_i = xv_{i,j}^{-1}W_i,$$

because k stabilizes W_i . There are finitely many choices of x, i, j , proving the claim. $\square$

We have obtained a nonzero stationary current and a conull family of unique bilateral trajectories. Every finite part of such a trajectory has a noncyclic stabilizer of the uniformly bounded total reduced rank from Lemma 3.3. The next section encodes these stabilizers by finite core graphs. Passing to the closed support there will use closedness of finite-graph liftability, rather than treating a conull assertion as a support assertion.

5 Finite carriers for a stationary current

We continue with the restricted HNN splitting, the free group P , and the nonzero atomless stationary current

$$i_*\nu = j_*\nu = \mu$$

provided by Proposition 4.5. Let T_P be the Cayley tree of the fixed basis of P , let R be its basis rose, and write $\Omega = \text{supp } \mu$. Propositions 4.7 and 4.8 give a conull set $\mathcal{U} \subset \Omega$ of lines with irrational endpoints, together with their unique aligned bilateral trajectories through the canonical edge e of the Bass–Serre tree $\mathcal{T}$. Transfers along finite parts of these trajectories preserve measure.

Our objective is to identify the finitely generated noncyclic subgroups that can fix an entire bilateral trajectory. The window stabilizers have bounded rank but can have arbitrarily large core graphs. We keep track of that distinction: after suppressing vertices of degree two, the graph sizes are bounded, and the edges whose labels remain bounded produce a finite family of subgroup carriers. Maximal carriers return to themselves under transfer, giving automorphisms to which we can apply Brinkmann’s theorem.

5.1 Window cores and their limits

We use the immersed-core description of finitely generated free subgroups and the corresponding pullback description of intersections; see Stallings [22, Section 3.3, Theorem 5.5, and Section 7]. The rank bounds and limiting carrier construction needed here are proved below.

For a finitely generated nontrivial subgroup $K \leq P$, put

$$C(K) = K \setminus \text{Hull}_{T_P}(\Lambda K).$$

This is its finite unbased core graph, with its label-preserving immersion into R . It has no vertices of degree one. A circuit in an immersed graph will mean a locally injective map from a subdivided circle; it need not be an embedded simple cycle.

For each $n \geq 1$, choose one representative of every P -orbit of positively traversed windows

$$W = (E_{-n}, \dots, E_n), \quad E_0 = e,$$

whose full pointwise stabilizer $K_W = \text{Stab}_P(W)$ is noncyclic. Define the finite graph

$$\Gamma_n = \coprod_W C(K_W), \quad \Gamma_n \longrightarrow R.$$

The choice is finite by Lemma 3.3. Set $r_0 = \bar{r}(P)$. Since μ is nonzero and atomless, P has rank at least two, and hence $r_0 \geq 1$.

A *state* of a labeled core is one of its vertices. In each noncyclic core, call a vertex *natural* if its degree is at least three. A *natural edge* is a maximal nonbacktracking path whose interior vertices have degree two. Suppressing those interiors gives the *natural graph*.

Lemma 5.1 (Window cores). *The following properties hold.*

1. *The sum of the reduced ranks of the components of Γ_n is at most r_0 . After suppressing degree-two interiors, Γ_n has at most $3r_0$ unoriented edges and $2r_0$ vertices.*
2. *Every line in Ω has a label-preserving lift to Γ_n .*
3. *A ray with irrational endpoint has at most one starting state for a lift among all components of Γ_n .*

The same statements hold for any other fixed finite positive window shape containing its specified central edge, with the same bounds.

Proof. The reduced-rank assertion is Lemma 3.3. In a noncyclic connected core, suppressing degree-two interiors leaves a graph all of whose vertices have degree at least three. If E, V are its edge and vertex counts and r its rank, then

$$2E \geq 3V, \quad E - V = r - 1.$$

Consequently $E \leq 3(r - 1)$ and $V \leq 2(r - 1)$. Sum these inequalities over the components. This bounds natural edges, not their subdivided lengths.

We describe precisely what a state of a core represents. If the representative window is W and $K = K_W$, a vertex Kx of $C(K)$ represents the actual window $x^{-1}W$ through e . This description does not depend on the chosen representative of the coset. It is injective: equality $x^{-1}W = y^{-1}W$ says $yx^{-1} \in \text{Stab}_P(W) = K$, or equivalently $Kx = Ky$. States in different representative cores cannot represent the same actual window, because that would identify their P -orbits.

A ray starting at Kx lifts in T_P starting at x and remaining in $\text{Hull}(\Lambda K)$. Its label, read from the identity, therefore has endpoint in $\Lambda(x^{-1}Kx)$. If its endpoint is irrational, Proposition 4.8 forbids two distinct represented windows carrying that endpoint. This proves uniqueness of the starting state. Continuation from a state is unique because the core map is an immersion.

For a line in $\mathcal{U}$, the endpoints lie in the limit set of its window stabilizer by Proposition 4.8; hence its labels lift to Γ_n . Liftability to a fixed finite graph is closed in line space. Indeed, choose a vertex on a limiting line. On each bounded centered segment, sufficiently close lines have the same labels. There are only finitely many possible lift states for the chosen vertex, so a subsequence uses one state for arbitrarily long centered segments. The immersion then gives a lift of the whole line from that state. Since $\mathcal{U}$ is dense in Ω , every Ω -line lifts. This is the only extension from the conull set to the entire support used here; we have not yet asserted that all support lines admit transfers.

The proof applies to any fixed window shape, using the corresponding rank bound and trajectory uniqueness. $\square$

We shall also use Lemma 4.9: for a fixed window shape and a fixed ball in T_P , only finitely many actual windows with noncyclic stabilizer have their stabilizer hull meeting that ball. In the preceding state description this is particularly transparent. If $x^{-1}\text{Hull}(\Lambda K)$ contains a vertex y of the ball, then Kxy is a vertex of the finite core $C(K)$. The finite choices of y and Kxy determine Kx , hence the actual window $x^{-1}W$.

Definition 5.2. Pass to a subsequence $n \rightarrow \infty$ on which the natural graph of Γ_n has a fixed abstract type, and identify those natural graphs with one fixed finite graph. Pass to a further subsequence on which each natural edge has either a fixed label or length tending to infinity. For every edge of the latter kind, arrange convergence of all finite prefixes of its label in both orientations. The *small graph* Γ^{sm} consists of the natural vertices and the natural edges with fixed labels, subdivided according to those labels.

Here is the order of these choices. There are only finitely many natural graph types with at most $3r_0$ edges and $2r_0$ vertices. For each of the finitely many edges of a chosen type, take a subsequence on which its lengths are bounded or tend to infinity. In the bounded case there are only finitely many possible reduced labels, so fix one. In the divergent case diagonalize over the prefix length, simultaneously in both orientations and on every such edge. These choices leave an infinite subsequence with $n \rightarrow \infty$. No bound on the lengths of the divergent edges is assumed.

Figure 1 illustrates the distinction between a fixed natural graph and its fixed small subgraph.

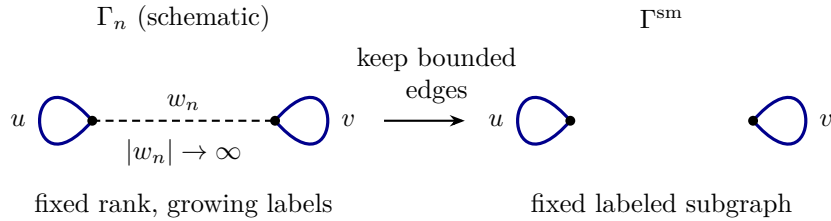


Figure 1: A fixed natural graph can have arbitrarily long edge labels. Here the bounded loops survive and the divergent bridge is deleted, while both natural vertices remain. The small components in this schematic example are cyclic; a noncyclic small component is not assumed to exist.

Each connected component D of Γ^{sm} , with a chosen vertex, gives the subgroup of P consisting of labels of its based loops. The map to R is an immersion, so this subgroup is isomorphic to $\pi_1 D$. Changing the vertex changes its conjugacy class only. Components of rank zero or one are allowed. A bi-infinite reduced path entirely in D lies in its unbased core, and its labels have endpoints in a translate of the corresponding subgroup limit set.

5.2 A finite cofinal family of subgroups

Let $\mathcal{C}$ be the set of P -conjugacy classes of finitely generated noncyclic subgroups $Q \leq P$ fixing a positively traversed bilateral path through e pointwise. Such a path is unique: otherwise its two finite windows would eventually diverge, contrary to Proposition 4.8. Define

$$[Q] \preceq [M] \iff Q \leq aMa^{-1} \text{ for some } a \in P.$$

We first verify that this really is a partial order, rather than merely a preorder.

Lemma 5.3. *A nontrivial finitely generated subgroup of a finite-rank free group cannot be properly contained in a conjugate of itself. Consequently $\preceq$ is a partial order on $\mathcal{C}$.*

Proof. Suppose $a^{-1}Ma \subsetneq M$. No positive power a^k belongs to M : otherwise the nested inclusions

$$M \supseteq a^{-1}Ma \supseteq \cdots \supseteq a^{-k}Ma^k = M$$

force equality at every stage. Thus the Schreier vertices Ma^{-n} , $n \geq 0$, are distinct. Fix a nontrivial reduced word $w \in M$, of length L . At each of those vertices the word w labels a closed reduced path, since $a^{-n}wa^n \in M$.

The Schreier graph consists of its finite unbased core with trees attached. A closed reduced path of length L has its starting vertex at distance at most $L/2$ from the core: outside it, the path must travel to the core and return along the same tree attachment. The $L/2$ -neighborhood of the finite core has only finitely many vertices. It cannot contain all the distinct Ma^{-n} , proving the first assertion. The reverse direction of conjugate containment follows by replacing a by a^{-1} .

If $[Q] \preceq [M] \preceq [Q]$, suitable representatives give

$$Q \leq aMa^{-1} \leq abQ(ab)^{-1}.$$

The inclusion between the first and last groups is equality by the first assertion. Both intermediate inclusions are therefore equalities, and $[Q] = [M]$. Reflexivity and transitivity are immediate. $\square$

The following elementary graph bound supplies the compactness step. It is important here that a core can have bridges.

Lemma 5.4 (Bounded circuits in a core). *Let C be a finite connected graph with no degree-one vertices and with at least one edge. Every edge of C lies on an immersed circuit of length at most $2|E(C)|$, where edges have length one and $|E(C)|$ counts unoriented edges.*

Proof. A nonbridge edge lies on a simple cycle of length at most $|E(C)|$. If a is a bridge, removing its interior leaves two components, each containing a cycle. Indeed, a component without cycles would be a finite tree with at most one possible vertex of degree one, which is impossible. On each side choose a shortest path from the endpoint of a to a simple cycle, traverse the cycle, and return along that path. The resulting based path is reduced and has length at most twice the number of edges on that side. Combine these two paths with a and a^{-1} . The joins are reduced, including the cyclic join, because the based paths remain on their respective sides of a . The resulting immersed circuit contains a and has length at most $2|E(C)|$. $\square$

Proposition 5.5 (Finite cofinal carriers). *The noncyclic component subgroup classes of Γ^{sm} belong to $\mathcal{C}$ and form a finite cofinal subset: every $[Q] \in \mathcal{C}$ is contained, up to P -conjugacy, in one of them. In particular $\mathcal{C}$ has only finitely many maximal elements, and they are exactly the maximal members of this finite family.*

Proof. Let M be a noncyclic subgroup given by loops at a chosen vertex of a small component. These same loop labels occur at the corresponding state of every selected Γ_n . By the state interpretation in Lemma 5.1, M fixes an actual centered window of radius n . The windows obtained for different n have identical restrictions to any common shape, because M is noncyclic. Their compatible restrictions form a bilateral positive path fixed by M . Thus $[M] \in \mathcal{C}$.

Now take $[Q] \in \mathcal{C}$, and let $C(Q)$ be its fixed unbased core. Put

$$D_Q = 2|E(C(Q))|.$$

For every selected n , the group Q is contained in the full stabilizer of its centered radius- n window. After translating to one of our representatives, this inclusion gives a label-preserving immersion

$$C(Q) \longrightarrow \Gamma_n.$$

One way to see the map directly is to include the minimal Q -invariant subtree in the minimal subtree of the containing group and take quotients. Conjugating the containing group merely translates its subtree and does not change labels.

Choose n so large that every divergent natural edge of Γ_n has length greater than D_Q . Suppose the image of some edge of $C(Q)$ meets the interior of one of those natural edges. By Lemma 5.4, that source edge lies on an immersed circuit of length at most D_Q . Its image remains immersed. At the degree-two interior of a natural edge there is no reduced way to turn around, so the image circuit traverses that entire natural edge. This remains true if its chosen starting point is in the interior: view the traversal cyclically. Its length is then greater than D_Q , a contradiction.

The whole image of $C(Q)$ therefore lies in Γ^{sm} . Since it is connected, it lies in one small component, with subgroup M . The induced inclusion gives $[Q] \preceq [M]$, and M is noncyclic because it contains a noncyclic subgroup. Notice that no compatible choices of conjugating elements were required as n varied: once all long edges exceed D_Q , the argument applies to every possible label-preserving immersion.

There are finitely many small components. A maximal member of their noncyclic subgroup classes is maximal in all of $\mathcal{C}$: any larger class lies below another member of the finite family and hence would contradict its maximality there. Conversely a maximal class in $\mathcal{C}$ equals a small class above it. This proves the final assertion. $\square$

We have now replaced the possibly unbounded window cores by a finite set of maximal subgroup carriers. The next step uses maximality to make a return transfer surjective. If there are no noncyclic small components, $\mathcal{C}$ is empty and all statements about its maximal members below are vacuous.

5.3 Return automorphisms

Moving one edge forward along the path fixed by Q and expressing Q in the new edge coordinates defines a map

$$F : \mathcal{C} \longrightarrow \mathcal{C}.$$

This moves subgroup coordinates, not individual support lines; it is defined for every class in $\mathcal{C}$.

Proposition 5.6 (Atoroidal return). *The map F is an order-preserving bijection with order-preserving inverse. It permutes the finite set of maximal classes of $\mathcal{C}$. For every representative M of such a class, there are an integer $p \geq 1$ and an element $v \in G$ of height p such that*

$$v^{-1}Mv = M.$$

The automorphism $\alpha : M \rightarrow M$, $\alpha(m) = v^{-1}mv$, is atoroidal: no nontrivial conjugacy class in M is periodic under α .

Proof. Choose a frame h for the next edge on the unique path fixed by Q . Transfer gives $h^{-1}Qh \leq P$. Another frame has the form hq , $q \in P$, and conjugates this output inside P . Conjugating Q inside P translates its path and gives the same output class. Thus F is well defined. Moving backward is its inverse. If $Q \leq M$ after choosing representatives, both groups fix the same path by noncyclic uniqueness; the same partial conjugation preserves their inclusion. This proves the order assertions.

An order automorphism permutes maximal elements. There are finitely many of them by Proposition 5.5, so $F^p[M] = [M]$ for some $p \geq 1$. If their number is N , one may choose $p \leq N$.

Let h be a frame for the p th forward edge. There exists $c \in P$ with

$$h^{-1}Mh = cMc^{-1}.$$

Set $v = hc$. Then v has height p and $v^{-1}Mv = M$, which proves that the return map is an onto automorphism.

Suppose $\alpha^k(w) = u^{-1}wu$ for some nontrivial $w \in M$, $u \in M$, and $k \geq 1$. Then

$$g = v^k u^{-1}$$

commutes with w . Its height is $kp \neq 0$, whereas w has height zero and infinite order. A relation $w^a g^b = 1$ forces $b = 0$ by height, then $a = 0$. Thus $\langle w, g \rangle \cong \mathbb{Z}^2 = BS(1, 1)$, contrary to BS-freeness. This proves atoroidality without using hyperbolicity of G or the flaring statement under investigation. $\square$

The element v preserves the bilateral path fixed by M : normalizing M sends that path to another path fixed by M . Both paths contain ve , so uniqueness in that edge's coordinates identifies them. It translates the path forward by p edges. In particular, the paths from e to $v^b e$ and to $v^{-b} e$ are the concatenated positive and negative return paths, respectively, and are reduced.

5.4 Expansion and vanishing carrier mass

For an element w of a finitely generated free group with a chosen basis, write $|w|$ for based word length and $\ell(w)$ for cyclic length. Brinkmann's theorem states that an automorphism α of a finite-rank free group with no nontrivial periodic conjugacy class is hyperbolic: there exist $b_0 \geq 1$ and $\lambda_0 > 1$ such that

$$\lambda_0 |w| \leq \max\{|\alpha^{b_0}(w)|, |\alpha^{-b_0}(w)|\} \quad \text{for every } w$$

[3, Theorem 1.1 and the definition of a hyperbolic automorphism]. These are based lengths in a fixed free basis. We need both a stronger expansion factor and a consequence for arbitrary currents; the latter uses a sum, not a maximum.

Lemma 5.7 (Expansion on elements and currents). *Let M, v, α be as in Proposition 5.6, and choose a free basis of M . There exist $b \geq 1$, an automorphism $\beta = \alpha^b$, and $\lambda > 4$ such that*

$$\lambda |w|_M \leq \max\{|\beta(w)|_M, |\beta^{-1}(w)|_M\}, \quad w \in M, \quad (5.1)$$

$$\lambda \ell_M(w) \leq \max\{\ell_M(\beta(w)), \ell_M(\beta^{-1}(w))\}, \quad w \in M. \quad (5.2)$$

For every M -current σ one has

$$\lambda \|\sigma\|_M \leq \|\beta_* \sigma\|_M + \|(\beta^{-1})_* \sigma\|_M. \quad (5.3)$$

The partial conjugations along the paths to $v^b e$ and $v^{-b} e$ have domains

$$S_+ = P \cap v^b P v^{-b}, \quad S_- = P \cap v^{-b} P v^b,$$

respectively. These are finitely generated full window stabilizers, and both contain M . Their boundary transfers are defined on all of $\partial^2 \Lambda S_+$ and $\partial^2 \Lambda S_-$. The transfers Θ_+, Θ_- restrict on $\partial^2 \Lambda M$ to β, β^{-1} , respectively, and preserve μ on their irrational-endpoint domains.

Proof. Apply Brinkmann's theorem to α , and put $\gamma = \alpha^{b_0}$. If $|\gamma(w)|_M \geq \lambda_0|w|_M$, applying the inequality to $\gamma(w)$ shows that its expanding direction must again be forward: its backward image has length $|w|_M$, which is strictly less than $\lambda_0|\gamma(w)|_M$ for $w \neq 1$. Induction gives

$$|\gamma^k(w)|_M \geq \lambda_0^k|w|_M.$$

The analogous statement holds for a backward initial expanding direction. Choose $k \geq 1$ with $\lambda_0^k > 4$, and set $b = b_0k$, $\beta = \alpha^b$, and $\lambda = \lambda_0^k$. This proves (5.1); it also holds for $w = 1$. Apply it to w^q , divide by q , and let $q \rightarrow \infty$. The identity $\lim_{q \rightarrow \infty} |w^q|_M/q = \ell_M(w)$ proves (5.2).

For a counting current η_w , the cyclic inequality and $\|\eta_w\|_M = \ell_M(w)$ imply (5.3), with the maximum replaced by the sum. The sum form is linear and therefore holds for all finite nonnegative linear combinations of counting currents. We give the approximation argument extending it to every current.

Fix $k \geq 2$. The directed graph with vertices the reduced words of length $k-1$ and edges the reduced words of length k has an edge from a word's prefix to its suffix. The masses $\langle u, \sigma \rangle$ of directed segment cylinders based at the identity and labeled by length- k words u form a nonnegative circulation: both incoming and outgoing sums at a vertex equal its length- $(k-1)$ cylinder mass. Any such circulation is a finite nonnegative sum of directed cycle circulations. Indeed, follow positive-flow edges to find a directed cycle, subtract the minimum edge flow on that cycle, and repeat; at least one positive edge disappears at each step.

Write the circulation as $\sum_i a_i C_i$. Each directed cycle is the block circulation of a cyclically reduced word w_i , including multiplicities if that word is a proper power. Let C_i^{flip} be the circulation obtained by reversing words and inverting letters. The counting current η_{w_i} has circulation $C_i + C_i^{\text{flip}}$. Since σ is flip-invariant, the current

$$\sigma_k = \frac{1}{2} \sum_i a_i \eta_{w_i}$$

has exactly the same masses as σ on all length- k cylinders. Summing over extensions gives equality on all shorter word cylinders as well. This also covers graph loops and period-one words; proper powers have their prescribed counting multiplicities.

As $k \rightarrow \infty$, the currents σ_k converge to σ on every cylinder, hence in the current topology. Their norms are equal to that of σ , since length-one masses already agree. Continuity of the norm and of the current maps induced by $\beta^{\pm 1}$ now proves (5.3).

Finally, the return paths are reduced as observed after Proposition 5.6. Fixing their first and last edges is equivalent to fixing the intervening segment, so their full stabilizers in P are exactly S_+ and S_- above. Since v normalizes M , both contain M . Their partial conjugations are $x \mapsto v^{-b}xv^b$ and $x \mapsto v^b xv^{-b}$, whose restrictions are β and β^{-1} . The boundary and measure assertions follow from Proposition 4.8. $\square$

Proposition 5.8 (Zero mass on a maximal carrier). *For every maximal carrier M supplied by Proposition 5.5,*

$$\mu(\partial^2 \Lambda M) = 0.$$

The same holds for every P -translate of this line set.

Proof. Use the boundary embedding of M in P to pull back the restriction of μ to the relatively closed subset $\partial^2 \Lambda M$ of $\partial^2 P$. This gives an M -current σ : invariance and flip-invariance come from μ , and the restriction is Radon because the boundary embedding sends compact subsets of $\partial^2 M$ to compact subsets of $\partial^2 P$.

Both endpoints are irrational outside a μ -null set by Proposition 4.7. On those lines the return transfer Θ_+ is measure-preserving, maps $\partial^2 \Lambda M$ onto itself, and agrees with β . Thus $\beta_* \sigma = \sigma$; the inverse has the same property. Equation (5.3) gives

$$\lambda \|\sigma\|_M \leq 2 \|\sigma\|_M.$$

Since $\lambda > 4$, this forces $\sigma = 0$. Hence the restricted measure is zero. Translation invariance of μ gives the last assertion. $\square$

This is a measure conclusion, not a support conclusion: a closed set of measure zero can still contain lines of Ω . The next section uses the return expansion to exclude precisely those possible support lines.

6 Expansion and periodic lines in the support

We retain the nonzero stationary current μ , its closed support $\Omega \subset \partial^2 P$, and the invariant conull set $\mathcal{U}$ from the preceding sections. Our purpose here is to exclude periodic lines from Ω . This is stronger than atomlessness: a line can belong to the support while having measure zero. Only after proving this stronger assertion will we extend the trajectory construction from $\mathcal{U}$ to every line of Ω .

The obstruction is a possible carrier line in the support despite its zero measure. We will expand a positive-measure cylinder near such a line along a fixed return. If the whole cylinder follows the return forever, its translated images concentrate on the zero-mass carrier set. If a line leaves the prescribed return domain, it follows a competing window; the cyclic intersection of the two window stabilizers forces a periodic middle in their long common segment. Lemma 6.3 records this alternative quantitatively. We first use it to produce periodic support from carrier support, then use whole-period pullback to rule out periodic support itself.

The window results provide the finite data for this argument. For each fixed shape there are finitely many representative cores of noncyclic window stabilizers; only finitely many actual windows have their hulls meeting a fixed ball. Distinct actual windows of that shape have at most cyclic common stabilizer, and an irrational endpoint determines at most one window (Proposition 4.8 and Lemma 4.9). We will also use the fixed small subgraph of the selected cores Γ_n from Definition 5.2.

Let $M \leq P$ represent one of the maximal noncyclic carrier classes. Proposition 5.6 and Lemma 5.7 supply an automorphism β of M , a number $\lambda > 4$, and fixed forward and backward window transfers Θ_+ and Θ_- with domains

$$\partial^2 \Lambda S_+ \quad \text{and} \quad \partial^2 \Lambda S_-, \quad M \leq S_+, S_- \leq P.$$

Here and below $\partial^2 \Lambda K$ denotes the ordered pairs of distinct points of the limit set of K . On the boundary of M these transfers are respectively β and β^{-1} . They are induced by the actual paths to the positive and negative powers of the return frame. In an M -basis,

$$\lambda |w|_M \leq \max\{|\beta(w)|_M, |\beta^{-1}(w)|_M\} \quad (w \in M). \quad (6.1)$$

The same inequality holds for cyclic lengths. Moreover, Proposition 5.8 gives

$$\mu(\partial^2 \Lambda M) = 0. \quad (6.2)$$

All transfers preserve μ on their domains in $\mathcal{U}$ and take those domains into $\mathcal{U}$. No assertion about their action on arbitrary support lines is being assumed.

We call a line of the P -Cayley tree an M -line if its two endpoints belong to ΛM . It corresponds to a unique line of the M -Cayley tree under the boundary embedding. A periodic line is the axis of a nontrivial element. Its primitive translation is a generator of its cyclic stabilizer; this does not mean a member of a free basis. When both M and P occur, we specify in which group the translation is primitive.

6.1 Uniform geometric estimates

Write $\mathcal{C}_Q$ for the unit-edge Cayley tree of a finitely generated free group Q with a fixed basis. A homomorphism on vertices is extended over each edge by its reduced image path. The image of a geodesic under a fixed injective homomorphism is a uniform quasigeodesic; its tightened line is at uniformly bounded Hausdorff distance from it. We first record the consequence needed both here and in the final argument.

Lemma 6.1 (Nearness and order under a fixed transfer). *Let $a : Q \rightarrow F$ and $b : Q \rightarrow F'$ be injective homomorphisms between finitely generated free groups with fixed bases. For every $D \geq 0$ there are constants $E, T \geq 0$ with the following properties. Let L be an oriented line of $\mathcal{C}_Q$, and let L_a, L_b be its tightened image lines in $\mathcal{C}_F, \mathcal{C}_{F'}$. If $q \in Q$ satisfies $d(a(q), L_a) \leq D$, then*

$$d(b(q), L_b) \leq E.$$

If two such elements q_1, q_2 have projections to L_a positively ordered and separated by more than T , then their projections to L_b are in the same order. As their separation on L_a tends to infinity, their separation on L_b tends to infinity, uniformly over all L, q_1, q_2 .

Proof. Fix Hausdorff bounds between each tightened image line and the images of vertices of L . Choose a vertex $u \in L$ whose a -image is uniformly close to a point of L_a closest to $a(q)$. The lower quasi-isometry bound for a gives a bound on $d_Q(q, u)$ depending only on D and a . The upper bound for b , followed by its line Hausdorff bound, proves the first assertion.

For the order assertion, first consider actual vertices $u < v$ on an oriented source line and one fixed embedding f . The image of v is within a fixed distance of the geodesic from $f(u)$ to $f(z)$ for every source vertex z sufficiently far beyond v . Projecting to the tightened image line and letting z tend to its positive end shows that the projection of $f(v)$ cannot lie more than a fixed distance behind that of $f(u)$. On the other hand, the distance between these projections is at least

$$c_f d_Q(u, v) - C_f$$

for fixed $c_f > 0, C_f \geq 0$, by the lower embedding bound and the line Hausdorff bound. Thus sufficiently separated vertices have projections in the correct order, with separation tending to infinity.

Apply this first to a and then to b . Replace q_i by vertices $u_i \in L$ at the uniformly bounded distance obtained in the first paragraph. Their image projections change by uniformly bounded amounts. Sufficient positive separation of the $a(q_i)$ therefore forces $u_1 < u_2$ with large separation, and then forces the asserted order and separation of the $b(q_i)$. All constants came from the fixed embeddings and D . $\square$

We will also use two elementary properties of trees. If x, y lie within D of a geodesic line L , the part of $[x, y]$ farther than D from both endpoints lies on L . Indeed, project x, y to L ; if the projections are distinct, the connecting segment enters and leaves L there. If they coincide, then $d(x, y) \leq 2D$, so there is no point farther than D from both endpoints. The same assertion holds

for an arbitrary convex subtree. Secondly, the lines meeting a fixed finite ball form a compact subset of line space: they are a finite union of the compact sets of lines through its vertices. A current has finite mass on this set.

Lemma 6.2 (Expansion with fixed trimming). *Fix a maximal carrier M and the maps above. There are constants $Q \geq 0$ and $L_* > 0$ with the following property. Let L be an oriented M -Cayley-tree line, let x, y be vertices on L with $d_M(x, y) = l \geq L_*$, and let $\mathcal{A} \subset \mathcal{U}$ be a Borel set whose lines lie within Q of both x, y in $\mathcal{C}_P$. Suppose a sign $\epsilon \in \{+1, -1\}$ satisfies*

$$d_M(\beta^\epsilon x, \beta^\epsilon y) \geq \lambda l.$$

If every line of $\mathcal{A}$ belongs to $\partial^2 \Lambda S_\epsilon$, then its image $\mathcal{A}' = \Theta_\epsilon(\mathcal{A})$ has the same measure, lies in $\mathcal{U}$, and admits vertices x', y' on the tightened line $\beta^\epsilon L$ such that

$$d_P(x', \ell), d_P(y', \ell) \leq Q \quad (\ell \in \mathcal{A}'), \quad d_M(x', y') \geq 2l.$$

The same sign satisfies

$$d_M(\beta^\epsilon x', \beta^\epsilon y') \geq \lambda d_M(x', y').$$

The constants are independent of $L, x, y, \mathcal{A}$ and of the number of times this assertion is applied.

Proof. Choose Q bounding the distance of every M -vertex on a geodesic segment or line to its tightened P -image, and bounding the reverse distance from that image to images of vertices. Increase it if necessary. Applying Lemma 6.1 to the two fixed window maps gives a common C_1 such that $u = \beta^\epsilon x$ and $z = \beta^\epsilon y$ are within C_1 of every line in $\mathcal{A}'$.

Choose lower embedding constants $a > 0, b \geq 0$ with

$$d_P(g, h) \geq ad_M(g, h) - b \quad (g, h \in M).$$

Let B be a common vertex-to-line bound for the two automorphisms $\beta^{\pm 1}$ of M . Choose an integer R such that

$$R > B, \quad aR - b - Q > C_1. \quad (6.3)$$

Delete R edges at both ends of $[u, z]$ in $\mathcal{C}_M$, and call its new endpoints x', y' . The first inequality in (6.3) puts them on the tightened line $\beta^\epsilon L$. Their P -images lie within Q of $[u, z]_P$. If v is either new endpoint, its projection to this segment has distance at least $aR - b - Q > C_1$ from each end. The intervening part of $[u, z]_P$ lies on every line of $\mathcal{A}'$, by the tree observation. Hence x', y' are still within Q of every such line.

Put

$$K = \max\{\text{Lip}(\beta), \text{Lip}(\beta^{-1})\}.$$

The new length l' satisfies

$$l' \geq \lambda l - 2R, \quad d_M(\beta^{-\epsilon} x', \beta^{-\epsilon} y') \leq l + 2KR. \quad (6.4)$$

Choose L_* sufficiently large that, for every $l \geq L_*$,

$$\lambda l - 2R \geq 2l, \quad l + 2KR < 2\lambda l.$$

Then the opposite direction at the new pair has length strictly less than $\lambda l'$. Inequality (6.1) forces expansion in the original direction. Measure preservation and membership in $\mathcal{U}$ follow from the invariant conull transfer construction. Since the maps are fixed boundary homeomorphisms on their domains, the image set is Borel. $\square$

6.2 A failed return produces a periodic segment

Fix the carrier M , its return maps, and the constants in Lemma 6.2. The next lemma supplies the common construction for both support arguments. Its reference line is transferred in the boundary of M ; only the competing line in its conclusion is asserted to lie in Ω .

Lemma 6.3 (Finite failure of an expanding return). *There are constants $L_0 \geq L_*$, $c > 0$, $C_2 \geq 0$, and an integer $D \geq 1$, depending only on the fixed splitting, carrier, and return maps, with the following property. Let $\eta \in \Omega \cap \partial^2 \Lambda M$, let L be its corresponding oriented M -Cayley-tree line, and choose vertices $x_0 < y_0$ on L with $l_0 = d_M(x_0, y_0) \geq L_0$. Suppose $\epsilon \in \{+1, -1\}$ satisfies*

$$d_M(\beta^\epsilon x_0, \beta^\epsilon y_0) \geq \lambda l_0.$$

Then there are $n \geq 0$, a line $\ell \in \mathcal{U}$, and an oriented periodic P -line ζ of primitive period at most D such that ℓ , ζ , and the tightened P -image η_n of $\beta^{\epsilon n} L$ share a segment of length at least

$$c 2^n l_0 - C_2. \tag{6.5}$$

The endpoints of ζ lie in the limit sets of the stabilizers of two distinct actual windows of the shape defining Θ_ϵ . One is the prescribed window, with stabilizer S_ϵ .

Proof. Choose $L_0 \geq L_*$ large enough that the projections of x_0, y_0 to η are distinct and occur in that order. This choice is uniform by Lemma 6.1; we will increase L_0 once more below. The cylinder prescribing the directed segment between those projections has positive finite measure, since $\eta \in \Omega$. Its intersection with $\mathcal{U}$ is a Borel set $\mathcal{A}_0$ of positive finite measure, every line of which is within Q of x_0, y_0 .

Attempt to apply Θ_ϵ successively to the entire current set. At time n , call the attempt a *failure* if $\mathcal{A}_n$ is not contained in $\partial^2 \Lambda S_\epsilon$. Otherwise define $\mathcal{A}_{n+1} = \Theta_\epsilon(\mathcal{A}_n)$ and use Lemma 6.2 to obtain the next segment. As long as the process continues, its vertices x_n, y_n lie on $\beta^{\epsilon n} L$, their distance l_n is at least $2^n l_0$, and every line of $\mathcal{A}_n$ lies within Q of both vertices. Its measure is the fixed number $\delta = \mu(\mathcal{A}_0) > 0$.

The process cannot continue forever. Choose an M -vertex a_n within one edge of the midpoint of $[x_n, y_n]_M$ and translate $\mathcal{A}_n$ by a_n^{-1} . The central portion of $[x_n, y_n]_P$ belongs to every line of $\mathcal{A}_n$. The image of a_n is uniformly close to this segment, and the lower embedding bound puts its projection arbitrarily far from both endpoints as n increases. That projection therefore lies in the common central portion. Thus the translated sets eventually lie in one compact set $\mathcal{K}$ of lines meeting a fixed ball. Their measures are still δ .

Every subsequential limit of these lines belongs to $\partial^2 \Lambda M$. To see this, take a subsequence of the centered M -segments. Both halves tend to infinite length, so a further subsequence converges to a bi-infinite M -geodesic. Its quasigeodesic image in P has two distinct boundary endpoints, and its tightened line is the limit of the growing common central segments. These endpoints lie in ΛM .

Consequently every relative neighborhood of $\mathcal{K} \cap \partial^2 \Lambda M$ in $\mathcal{K}$ contains all sufficiently late translated sets. Otherwise points outside one such neighborhood would have a contrary subsequential limit in the compact set $\mathcal{K}$. Outer regularity of the finite measure on $\mathcal{K}$ now gives

$$\mu(\mathcal{K} \cap \partial^2 \Lambda M) \geq \delta,$$

contrary to (6.2).

A failure produces a bounded-period axis. Suppose failure occurs at time n , and choose $\ell \in \mathcal{A}_n \setminus \partial^2 \Lambda S_\epsilon$. This line has an actual trajectory for the same finite one-sided window. Its stabilizer

K is noncyclic, and this actual path differs from the prescribed path: equality would put both endpoints of ℓ in ΛS_ϵ .

Both ℓ and the tightened image η_n of $\beta^{\epsilon n}L$ pass within Q of x_n, y_n . Since $x_n, y_n \in M \leq S_\epsilon$, their distances from $\text{Hull}(\Lambda S_\epsilon)$ equal the fixed number $d_P(1, \text{Hull}(\Lambda S_\epsilon))$. Set

$$H_\epsilon = \max\{Q, d_P(1, \text{Hull}(\Lambda S_\epsilon))\}.$$

Deleting H_ϵ from both ends of $[x_n, y_n]_P$ leaves a segment in ℓ , in η_n , and in the prescribed stabilizer hull. It also lies in $\text{Hull}(\Lambda K)$ because ℓ does. Its length is at least

$$al_n - C, \tag{6.6}$$

where $a > 0, C \geq 0$ depend only on the fixed maps and embeddings. In particular, no constant from the varying competing subgroup K is needed.

Read this segment in the two finite stabilizer cores. Their fiber product pairs core states, with an edge for each pair of equally labeled edges. At an actual vertex z the corresponding pair of states represents the two paths translated by z^{-1} . They remain distinct. Every loop in the corresponding component of the fiber product therefore fixes both paths, so its fundamental group injects into an at-most-cyclic group, by Proposition 4.8. The component has rank at most one. Since the window is fixed, there are only finitely many core representatives, and the numbers of unoriented edges in all relevant fiber products are bounded by one integer $D \geq 1$. Conjugating a stabilizer preserves its labeled unbased core and changes only the state used for this reading.

Increase L_0 so that $aL_0 - C > 2D + 1$, taking the maximum of the constants for the two signs. This ensures that every failure reading is long enough to contain a circuit. A reduced path in a finite tree has length at most its number of edges. A connected rank-one finite graph consists of one circuit with trees attached, so a sufficiently long reduced path follows that circuit between initial and terminal pieces of length at most D . Our reading therefore has a central periodic portion of length at least $al_n - C - 2D$. Extend it in the absolute Cayley tree to an oriented periodic line ζ . Its primitive P -period is at most D . Unfolding the circuit in each core puts its endpoints in both actual window stabilizer limit sets. Since $l_n \geq 2^n l_0$, this proves (6.5), with constants enlarged uniformly over the two signs. The line ℓ belongs to $\mathcal{U}$, as required. $\square$

The failure need not occur on a set of positive measure. Its chosen line already belongs to $\mathcal{U} \subset \Omega$, which is enough for the support limit below. Positive mass rules out indefinite transfer of the whole cylinder. Figure 2 shows how the two actual windows produce the periodic middle.

Proposition 6.4. *If Ω contains a line carried, up to translation, by a maximal noncyclic carrier M , then Ω contains a periodic line.*

Proof. Translate the given line into $\partial^2 \Lambda M$. Choose successively longer segments on its corresponding M -line and, for each segment, a sign supplied by (6.1). Apply Lemma 6.3, passing to a subsequence with the same sign. Its support line and bounded-period axis share a segment whose length tends to infinity.

Recenter this shared segment by an M -translation chosen near its midpoint on η_n . The translated periodic axes have period at most D and meet a fixed ball. There are only finitely many such axes: there are only finitely many cyclic labels of length at most D , and finitely many ways for one of their axes to meet that ball. Along a subsequence the periodic axis is fixed. The translated lines $\ell \in \Omega$ agree with it on segments tending to infinity in both directions, so they converge to it. Closedness of Ω proves that this periodic line belongs to Ω . $\square$

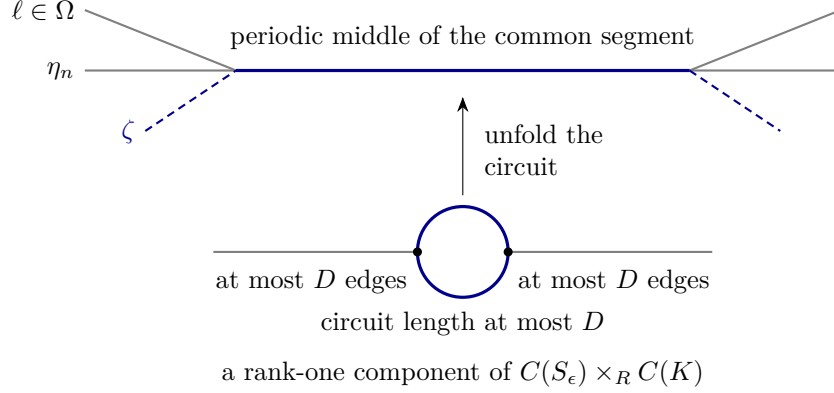


Figure 2: At a failed return, η_n and a competing support line ℓ share a long segment in two window-stabilizer hulls. Its reading in the core fiber product lies in a rank-one component, shown below. The initial and terminal tree portions have bounded length; between them the reading follows the circuit. Unfolding that circuit gives the periodic middle and its extension ζ . The figure is schematic; only ℓ is asserted to lie in the support.

6.3 Pulling back whole periods

The next lemma controls the number of periods in a common segment. Controlling only its metric length would be insufficient, since a period can grow under an inverse automorphism.

Lemma 6.5 (Bounded loss of periods). *Let ψ be an automorphism of a finitely generated free group M with a fixed basis. There is an integer $C_4 \geq 0$, depending only on ψ , with the following property. Suppose two lines L_1, L_2 of $\mathcal{C}_M$ share a segment $[u, z^k u]$, where u is a vertex, z translates L_2 , and $k > C_4$. Their tightened ψ -images share a segment consisting of $k - C_4$ whole $\psi(z)$ -periods. If z is primitive as a translation in M , then so is $\psi(z)$.*

Proof. Choose an integer $B \geq 0$ such that the image of any vertex on any line is within B of the tightened image line. Such a uniform constant exists by the quasigeodesic line bound for the fixed map ψ . Put $q = 2B + 1$ and $C_4 = 2q$.

Write $L'_i = \psi L_i$ for the tightened images and $z' = \psi(z)$. The line L'_2 is the axis of z' . If a is the projection of $\psi(u)$ onto this axis, the projection of the other endpoint image is exactly

$$\pi_{L'_2}(\psi(z^k u)) = \pi_{L'_2}(z'^k \psi(u)) = z'^k a,$$

because projection onto an invariant line commutes with its translation. Both endpoint images lie within B of both output lines. Thus $a, z'^k a$ lie within $2B$ of L'_1 .

The translation length of z' is an integer at least one. The vertices $z'^q a$ and $z'^{k-q} a$ therefore lie farther than $2B$ from both ends of $[a, z'^k a]$ and occur in the stated order. The tree observation puts their joining segment on L'_1 , and it already lies on L'_2 . It has exactly $k - 2q = k - C_4$ full periods, with both endpoints in the same integer z' -orbit. There is no additional phase loss. Automorphisms preserve being a non-power, proving the final assertion. $\square$

Lemma 6.6 (Agreement of periodic labels). *Two bi-infinite label sequences with respective positive integer periods a, b that agree on $a + b$ consecutive positions agree everywhere. Consequently two oriented periodic lines in a labeled Cayley tree with such a common segment are the same actual line.*

Proof. Number the common positions $0, \dots, a+b-1$. For $0 \leq i < a$, the letter at position i equals the letter at $i+b$ by the second period. By the first period this equals the letter at $(i+b) \bmod a$. Thus the first period pattern is invariant under addition by b modulo a , and is periodic with period $d = \gcd(a, b)$. The first b common letters then show that the second pattern is the same d -periodic pattern. Both entire sequences therefore agree. A label and an initial vertex determine a unique path in a Cayley tree, so equality of the sequences and the shared segment identifies the actual lines. $\square$

Lemma 6.7 (A bounded-period axis near a carrier). *Let $M \leq P$ be a finitely generated nontrivial subgroup, with fixed bases, and fix $D \geq 1$. There are constants L_D, B_D such that if a periodic P -line ζ of primitive period at most D shares a segment of length at least L_D with an M -line, then ζ is itself an M -line. Its primitive translation in M has cyclic M -length at most B_D .*

Proof. Let V be the number of vertices in the finite core of M in the P -rose, and take $L_D = (V+1)D$. The common segment lies in $\text{Hull}(\Lambda M)$. If r is the actual primitive P -translation of ζ , that segment contains $V+1$ successive vertices $r^i z_0$ in one period phase. Two have the same core state:

$$Mr^i z_0 = Mr^j z_0 \quad (0 \leq i < j \leq V).$$

Cancelling z_0 gives $r^{j-i} \in M$. Thus a power of the translation of this very axis belongs to M , and the whole line has endpoints in ΛM .

If z is its primitive translation in M , it is the smallest positive power of r belonging to M , up to orientation. Hence $z = r^q$ for some $1 \leq q \leq V$. Choose embedding constants with

$$|g|_M \leq A|g|_P + B \quad (g \in M).$$

Apply this to powers of z , divide by the exponent, and take the limit to obtain

$$\ell_M(z) \leq A\ell_P(z) \leq AVD. \tag{6.7}$$

Thus $B_D = AVD$ suffices. The two primitive translations need not coincide, which is why both groups are specified. $\square$

6.4 Exclusion of periodic support lines

Proposition 6.8. *The support Ω contains no periodic line.*

Proof. Suppose $\eta_0 \in \Omega$ is periodic.

Step 1: construct a trajectory and locate a maximal carrier. We first construct one compatible trajectory for η_0 ; uniqueness at its rational endpoints is not assumed.

Approximate η_0 by lines of $\mathcal{U}$. For each fixed centered window, their stabilizer hulls meet a common finite ball. Lemma 4.9 lets us pass to a subsequence with a constant actual window path. Taking these subsequences successively gives a bilateral aligned path whose every finite window has noncyclic stabilizer. The limiting endpoints belong to each such stabilizer limit set. Choose frames h_s on this path, with $h_0 = 1$. Boundary transfer on a fixed window is continuous, so the corresponding conull image lines converge in the successive frames to lines $\eta_s \in \Omega$. They are periodic: the core-state repetition argument in Lemma 6.7 puts a power of the original axis translation in each fixed window stabilizer, and this power is transferred to a nontrivial element translating the image line.

Choose the primitive P -translation r_s of η_s toward its positive endpoint. Write a forward step as $h_{s+1} = h_s p_s t q_s$, with $p_s, q_s \in P$. There is a least positive integer m_s with $r_s^{m_s} \in p_s I p_s^{-1}$. Its image is $r_{s+1}^{n_s}$, where n_s is the least positive integer with $r_{s+1}^{n_s} \in q_s^{-1} J q_s$. Minimality on the second side follows by applying the inverse subgroup isomorphism to its full axis stabilizer. The exponents are positive because we oriented each translation toward the transferred positive endpoint.

Recall the finite ambiguity dictionary of Lemma 4.2: if either minimum exponent exceeds one, or if one of these axes belongs to two distinct domain cosets on its side, one of its primitive root classes lies in a fixed finite set, up to conjugacy and inversion. Infinitely many such indices would repeat a root class along this actual reduced path. Lemma 4.3 would give a tree-hyperbolic element commensurating a nontrivial elliptic cyclic group, contrary to BS-freeness. Therefore only finitely many indices in the entire bilateral trajectory are exceptional in either way.

It follows that one can choose positive integers k_s for all $s \in \mathbb{Z}$ with

$$m_s \mid k_s, \quad k_{s+1} = n_s k_s / m_s. \quad (6.8)$$

Only finitely many ratios differ from one. Choose k_0 divisible by all the finitely many factors needed when propagating these relations in both directions. Then

$$h_s r_s^{k_s} h_s^{-1}$$

is one fixed nontrivial element, independent of s .

Fix s temporarily. The cyclically tightened circuit representing $r_s^{k_s}$ lifts closed to a component of each centered window core Γ_n , after recentering the trajectory at $h_s e$: the edges $E'_j = h_s^{-1} E_{s+j}$ have $E'_0 = e$ and so belong to the same family of windows defining Γ_n . Its length is fixed as the window size tends to infinity. It cannot traverse a natural edge whose length tends to infinity: any immersed closed path using the interior of such an edge traverses the whole edge, cyclically. Thus η_s is carried by the fixed small subgraph. If infinitely many positive times were carried only by cyclic small components, their primitive P -roots would again belong to finitely many conjugacy classes and repeat. We may therefore choose a time beyond every exceptional forward index at which the periodic line is carried by a noncyclic small subgroup. Enlarge it to a maximal carrier M and translate the line into $\partial^2 \Lambda M$. If there were no maximal noncyclic carrier, this conclusion itself would be a contradiction.

Step 2: follow expanding return iterates. The subsequent forward path agrees with the path fixed by M . At its first edge, both choices allow the periodic endpoints, and the domain coset is unique at this late time. Transfer and repeat to get agreement at every forward edge. In particular all positive β -iterates of the line belong to Ω and have unique forward choices along the prescribed return windows. This conclusion comes from the constructed trajectory, not from an as yet unproved support-invariance assertion.

Let $a_j \geq 1$ be the primitive cyclic M -length of its j th positive β -iterate. At some time one has $a_{j+1} \geq \lambda a_j$. Otherwise the cyclic version of (6.1), applied at every $j \geq 1$, would give $a_{j-1} \geq \lambda a_j$, impossible for positive integer lengths. Once a forward expansion occurs, it continues: the backward neighbor is too short to supply the next expanding alternative. Restart at such a time, and call the initial primitive period length d_0 .

Let z be the positive primitive M -translation of this initial line. Choose a vertex x_0 on its axis and $y_0 = z^k x_0$, where k is arbitrarily large. Then $l_0 = k d_0$. The cyclic expansion gives $\ell_M(\beta z) \geq \lambda d_0$, and therefore

$$d_M(\beta x_0, \beta y_0) = d_M(\beta x_0, (\beta z)^k \beta x_0) \geq k \ell_M(\beta z) \geq \lambda l_0.$$

This verifies the based-segment hypothesis even if βx_0 is off the axis of βz . For $l_0 \geq L_0$, Lemma 6.3 now supplies a failure time n , a support line, and a bounded-period axis sharing (6.5) with the n th return image.

Step 3: pull a failure axis back to the initial period. The resulting periodic P -line ζ has uniformly bounded period, allows both the prescribed and a competing path, and shares with the current M -line a segment of length at least (6.5). Take l_0 large enough that (6.5) exceeds the fixed threshold in Lemma 6.7, already at $n = 0$. That lemma implies that ζ is itself an M -line, with primitive M -period bounded by (6.7).

The two corresponding M -lines share a segment of length at least $c_0 2^n l_0 - C_0$, for fixed $c_0 > 0, C_0 \geq 0$. Here are the details of this change of metric. Near each end of their common P -segment, choose a vertex on each M -line whose P -image is within the fixed line Hausdorff bound. At each end the two chosen vertices are uniformly close in M , by the lower embedding estimate. The vertices chosen near opposite ends are separated in M by a positive multiple of the P -length minus a constant, by the upper embedding estimate. Choose the pairs in their oriented order, as justified by Lemma 6.1. In a tree, the two joining segments therefore have a common central segment after deleting a fixed amount at both ends. This gives the asserted lower bound.

The uniformly bounded primitive period of ζ now shows that this common segment contains at least

$$c' 2^n l_0 - C_3 \tag{6.9}$$

whole primitive ζ -periods, for fixed $c' > 0, C_3 \geq 0$. Apply Lemma 6.5 with $\psi = \beta^{-1}$ successively n times. At time zero at least

$$c' 2^n l_0 - C_3 - n C_4 \tag{6.10}$$

whole periods remain. Choose the initial length so large that

$$c' l_0 > \max\{C_4, C_3 + d_0 + 1\}. \tag{6.11}$$

Since $2^n \geq n + 1$, expression (6.10) exceeds $d_0 + 1$ for every possible failure time n . In particular all intermediate applications of the pullback lemma have enough periods.

The primitive period of the second line after these pullbacks may be large; call it $b \geq 1$. The number of full periods, rather than its metric length, was controlled. More than $d_0 + 1$ periods give a common label segment of length at least $(d_0 + 2)b \geq d_0 + b$. Lemma 6.6 identifies the original periodic line and the pulled-back ζ as the same actual M -line. Applying β^n identifies their lines at failure, in both the M - and P -trees. This line allows both the prescribed and the competing forward window. That contradicts the unique forward coset choices on the late periodic trajectory. Failure is impossible as well, completing the contradiction. $\square$

6.5 The trajectory construction on the entire support

We have excluded periodic support points without assuming that all support points admit unique transfers. We can now prove precisely that extension, together with the uniform bound needed to count special rays.

Corollary 6.9. *The following assertions hold for the closed support Ω .*

1. *No line of Ω is carried by the small subgraph. There is an integer B_0 such that every segment of a lift of an Ω -line to any of the selected subsequence cores Γ_n , if contained entirely in the small subgraph, has length at most B_0 .*

2. *Every endpoint of every line of Ω is irrational.*

3. *Every line of Ω lifts to each centered core union Γ_n and has a unique compatible bilateral trajectory. On each fixed window, the partial boundary transfer takes every support line in its domain to a line of Ω .*

Proof. An immersed bi-infinite path in a finite cyclic small component is periodic, and such a line is excluded by Proposition 6.8. A line carried by a noncyclic small component is carried, up to conjugacy, by a maximal noncyclic carrier. Proposition 6.4 would then produce a periodic support line, also impossible. A tree component cannot carry an immersed bi-infinite path. This proves the first assertion about whole lines.

If small-only segments of lifted support lines had unbounded length, center progressively longer such segments at an actual vertex and translate that vertex to the identity in P . The translated lines remain in Ω and lie in the compact set of lines through the identity. Since the small graph is fixed and finite, a subsequence of the centered lifted segments converges to an immersed bi-infinite path in it. The corresponding limit support line would be carried by the small graph. This contradiction gives the uniform bound B_0 .

A rational endpoint is represented by an eventually periodic ray. If such a ray lay on a support line, translating farther and farther along its periodic tail would give a periodic line as a limit of support lines. Closedness of Ω and Proposition 6.8 exclude this. Both endpoints of every support line are therefore irrational.

The existence of lifts to each finite core was already proved in Lemma 5.1: it follows from conull lifts, density, and closedness of liftability into a fixed finite graph. Each such lift gives an actual window path carrying the line's endpoints. The irrational-endpoint uniqueness in Proposition 4.8 makes these paths compatible when a window is shortened. They therefore give a unique bilateral trajectory.

It remains to check the support of transferred lines. Approximate a given support line by lines of $\mathcal{U}$ and fix one window. All approximants meet a common fixed ball, so the hulls of their window stabilizers do too. There are only finitely many possible actual paths. Any path occurring infinitely often carries the limiting endpoints, because its limit set is closed; irrational uniqueness identifies it with the limiting line's own path. Thus the approximants eventually have that same path. The boundary transfer on its fixed subgroup is continuous. Their images belong to $\mathcal{U} \subset \Omega$ and converge to the transferred limiting line, which is in Ω by closedness. This proves the final assertion without discarding any support points. $\square$

7 Special rays and the end of the combination proof

We finish the proof of Theorem 3.1. Suppose that G is not hyperbolic. Propositions 3.12 and 4.5 then provide a nonzero stationary current μ , its support Ω , and a conull invariant set $\mathcal{U}$. By Corollary 6.9, every endpoint of every line of Ω is irrational, and every support line has a unique bilateral trajectory. Each fixed finite-path transfer maps its domain in Ω into Ω . Moreover, the cores Γ_n have a fixed small subgraph along the chosen subsequence, and there is a uniform bound on the length of a segment of an Ω -line that lifts entirely to that small subgraph.

The remaining argument detects branching in the language of Ω . Bounded natural graph complexity leaves only finitely many rays with more than one possible past. A periodic orbit among their endpoints gives a partial conjugation moving points progressively farther along one ray. Its invariant measure cannot be finite on the lines through a vertex.

7.1 Finitely many rays with two pasts

Fix the basis of P used above, and write T_P for its Cayley tree. Orient each line from its negative endpoint to its positive endpoint. An infinite reduced word is *occurring* if it labels a positive ray in a line of Ω . An occurring word ρ is *left-special* if there are distinct basis letters a, b such that both $a\rho$ and $b\rho$ occur. These extensions are required to be reduced. Translating their rays to start at 1 gives two lines of Ω with a common positive ray and different incoming edges.

Lemma 7.1. *The set of left-special occurring words is finite and nonempty.*

Proof. Let ρ be left-special. Both of its predecessor lines lift to each Γ_n by Lemma 5.1. Their common endpoint is irrational, so the uniqueness of a starting state for an irrational ray puts both lifts at the same vertex. This vertex has two different incoming edges as well as the outgoing ray edge. It is therefore a natural vertex.

Let B bound the lengths of small-only segments of support lines, as in Corollary 6.9. Starting at this natural vertex, the lift of ρ follows at most B labelled edges in the small subgraph before entering a long natural edge. There are only finitely many possible prefix words of length at most B and finitely many oriented natural edge types. On a subsequence the prefix and the oriented type are fixed. The lengths of these long edges tend to infinity, and their oriented labels have a fixed infinite prefix limit by the construction of Γ_n . Thus ρ is the concatenation of one of finitely many prefixes and one of finitely many prescribed infinite words. This proves finiteness.

Suppose now that no occurring ray is left-special. Then there is an integer N such that every occurring word of length N has a unique predecessor letter. Indeed, otherwise arbitrarily long finite words would have two distinct predecessor letters. Passing to a fixed pair of letters and a convergent subsequence of their full-line extensions would give a left-special infinite word. This uses compactness of the space of lines through a vertex and the closedness of Ω .

Form the finite directed graph whose vertices are occurring words of length N and whose edges are occurring words of length $N + 1$, an edge running from its length- N prefix to its length- N suffix. Every vertex has indegree one and outdegree at least one. The sum of the indegrees is the number of vertices, so every outdegree is also one. The graph is a disjoint union of directed cycles. Every line in Ω would therefore have a periodic label, contrary to Proposition 6.8. $\square$

7.2 A return map with positive lag

Let $\mathcal{E}$ be the set of P -orbits of the positive endpoints of left-special rays. This is a finite nonempty set. The forward path transfer induces a permutation of $\mathcal{E}$. To see this, take two predecessor lines of such a ray. Their common irrational endpoint puts them in the same unique domain coset. The boundary embedding on that coset sends them to two support lines with the same positive endpoint and distinct negative endpoints. After their positive rays merge, the two incoming edges exhibit a left-special ray. A change of the input representative by an element of P only changes the output by an element of P . The backward transfer is the inverse construction, so the resulting map on $\mathcal{E}$ is a permutation.

Choose a left-special word based at 1 whose endpoint ξ represents a periodic element of $\mathcal{E}$, of period $p > 0$. Compose the p forward transfers and adjust the final frame by an element of P . The actual path construction gives an element $v \in G$ with nonzero height p and a partial conjugation

$$\Theta : S \longrightarrow T, \quad \Theta(x) = v^{-1}xv, \quad S = P \cap vPv^{-1}, \quad T = P \cap v^{-1}Pv, \quad (7.1)$$

whose boundary map fixes ξ . Here S and T are the full stabilizers, expressed in their initial and final frames, of the corresponding finite path. In particular they are finitely generated subgroups of P . The map and its inverse preserve μ on their respective domains in $\mathcal{U}$.

Write $r_0 = 1, r_1, r_2, \dots$ for the consecutive vertices of the ray to ξ in T_P . Since $\xi \in \Lambda S$ and S is finitely generated, choose $s_n \in S$ and a fixed $K \geq 0$ such that $d(s_n, r_n) \leq K$ for every n . The embedding Θ is a quasi-isometry onto T and fixes the boundary endpoint ξ . Consequently there is a sequence of nonnegative integers $f(n)$ with

$$d(\Theta(s_n), r_{f(n)}) \leq C, \quad f(n) \longrightarrow \infty, \quad |f(n+1) - f(n)| \leq C, \quad (7.2)$$

after increasing a fixed constant C . These are the ray versions of the transport estimates in Lemma 6.1.

Lemma 7.2. *After replacing Θ by its inverse if necessary, the choices above may be made so that*

$$f(n) - n \longrightarrow +\infty.$$

Proof. If $|f(n) - n|$ were bounded along an infinite subsequence, then $s_n^{-1}\Theta(s_n)$ takes only finitely many values on that subsequence. The elements s_n escape every finite ball. Hence there are distinct s_n, s_m and $d \in P$ with $\Theta(s_n) = s_n d$ and $\Theta(s_m) = s_m d$. It follows that

$$\Theta(s_n s_m^{-1}) = (s_n d)(s_m d)^{-1} = s_n s_m^{-1}.$$

The nontrivial element $s_n s_m^{-1}$ has infinite order and commutes with v . Their heights are respectively zero and $p \neq 0$, so they generate a subgroup isomorphic to $\mathbb{Z}^2$. This contradicts the Baumslag–Solitar exclusion.

Thus $|f(n) - n| \rightarrow \infty$. The signed sequence $f(n) - n$ has bounded successive increments by (7.2), so it is eventually either positive or negative. The positive case is the assertion. In the negative case choose $t_m \in T$ uniformly close to r_m and integers $g(m)$ such that $\Theta^{-1}(t_m)$ is uniformly close to $r_{g(m)}$. Because f tends to infinity with bounded increments, a first-crossing choice $n(m)$ satisfies

$$|f(n(m)) - m| \leq C, \quad n(m) \longrightarrow \infty.$$

Then t_m and $\Theta(s_{n(m)})$ are uniformly close. Applying the inverse quasi-isometry gives

$$g(m) = n(m) + O(1), \quad g(m) - m = n(m) - f(n(m)) + O(1) \longrightarrow +\infty.$$

This proves the assertion for the inverse map. □

7.3 A cylinder that cannot lose mass

After the possible inversion, use Θ, S, T, f for the resulting map and ray data, and K, C for their uniform approximation bounds. The labels of distinct suffixes of the ray (r_n) are distinct: an equality would make the ray eventually periodic and ξ rational. By Lemma 7.1, there is n_0 such that no suffix beginning after r_{n_0} is left-special.

Fix $s > n_0$. Any support line containing the entire directed suffix $[r_s, \xi)$ has its preceding edge uniquely determined, then the edge before that, and so on back to r_{n_0} . Moreover, its actual path for the fixed window defining Θ is the same as the path for ξ : an irrational endpoint determines that path uniquely. These two statements have the following finite form.

Lemma 7.3. *For each fixed $s > n_0$, there is N_s such that, whenever $N \geq N_s$, every line of Ω containing the directed segment $[r_s, r_N]$ contains r_{n_0} and belongs to the domain of Θ .*

Proof. For the first assertion, a contrary sequence has a convergent subsequence in the compact space of directed lines through r_s . The limit belongs to Ω and contains the whole suffix $[r_s, \xi)$. The condition of not containing the fixed finite segment $[r_{n_0}, r_s]$ is closed in this compact space. The limit would therefore contradict the unique-past property just proved.

For the domain assertion, each competing line has an actual path for the fixed window. Its stabilizer hull contains that line, hence meets the vertex r_s . There are only finitely many such actual paths by Lemma 4.9. A contrary sequence therefore has a subsequence with one fixed competing path. Its stabilizer limit set contains the limiting positive endpoint ξ . Uniqueness for an irrational endpoint identifies this path with the path defining Θ , which is a contradiction. $\square$

For $N \geq N_s$, let

$$U_N = \{\ell \in \mathcal{U} : \ell \text{ contains the directed segment } [r_s, r_N]\}.$$

This set has finite positive μ -measure: the cylinder is compact and open, meets the support, and $\mathcal{U}$ is conull. Its lines contain r_{n_0} and lie in the domain of Θ by Lemma 7.3. The elements s_{n_0} and s_N lie within K of these lines. Lemma 6.1, applied to the fixed domain inclusion $S \hookrightarrow P$ and the fixed injection $\Theta : S \rightarrow P$, gives a constant E such that their images lie within E of every line of $\Theta(U_N)$. Equation (7.2) then puts $r_{f(n_0)}$ and $r_{f(N)}$ within $E + C$ of those image lines. The input projections are in their positive order with separation at least $N - n_0 - 2K$. For sufficiently large N , the order and separation assertions of Lemma 6.1 put the two image projections, and then the two ray-vertex projections, in the same positive order.

Choose a fixed integer $D > E + C$. The tree observation preceding Lemma 6.2 shows that every line of $\Theta(U_N)$ contains the directed central segment $[r_{f(n_0)+D}, r_{f(N)-D}]$ for all sufficiently large N . Both E and D depend only on the fixed map and the uniform approximation bounds; neither depends on s or N .

Now choose $s > \max\{n_0, f(n_0) + D\}$, and then take N beyond N_s and the preceding transport thresholds. Lemma 7.2 also gives $f(N) - D \geq N + 1$ for every sufficiently large N . The invariance of $\mathcal{U}$ under the path transfers now yields

$$\Theta(U_N) \subseteq U_{N+1} \subseteq U_N. \tag{7.3}$$

Because Θ preserves measure on its domain, these inclusions give

$$\mu(U_N) = \mu(\Theta(U_N)) \leq \mu(U_{N+1}) \leq \mu(U_N).$$

All sufficiently late U_N have the same positive finite measure. Continuity from above shows that their intersection U_∞ has that same measure. Every line in U_∞ contains the full directed ray $[r_s, \xi)$.

For each $n \geq s$, translate U_∞ by r_n^{-1} . These sets all consist of lines through 1, and they all have the same positive measure. Their positive endpoints $r_n^{-1}\xi$ are pairwise distinct: an equality for $n \neq m$ would give a nontrivial element of P fixing ξ , making it rational. The translated sets are thus pairwise disjoint. This contradicts the finite mass of the compact set of lines through 1.

We have contradicted the existence of the nonzero stationary current forced by nonhyperbolicity. Therefore G is hyperbolic. This completes the proof of Theorem 3.1.

8 The Magnus induction

We now apply Theorem 3.1 to one-relator groups. The geometric form of the Magnus construction gives an HNN extension whose vertex group has smaller complexity and whose edge groups are

simultaneous Magnus subgroups. The induction will establish hyperbolicity. At each stage, Linton's theorem on Magnus subgroups of hyperbolic one-relator groups supplies the quasiconvexity needed by the combination theorem.

8.1 Graph presentations and Magnus subgroups

Let Γ be a finite connected graph, and let $\lambda: S^1 \rightarrow \Gamma$ be a nontrivial immersed combinatorial circuit. Write

$$X = \Gamma \cup_{\lambda} D^2, \quad S_{\lambda} = \lambda(S^1).$$

We call a connected subgraph $\Delta \subset \Gamma$ a *Magnus subgraph* if $S_{\lambda} \not\subset \Delta$; equivalently, Δ omits an edge traversed by λ . The following observation relates this definition to the usual generator-subset definition.

Lemma 8.1. *Let $X = \Gamma \cup_{\lambda} D^2$ be as above, and let $\Delta \subset \Gamma$ be a Magnus subgraph. There is a free basis Σ of $\pi_1 \Gamma$ in which $\pi_1 \Delta$ is generated by a subset of Σ omitting a letter occurring in the cyclically reduced relator. Consequently, $\pi_1 \Delta \rightarrow \pi_1 X$ is injective. Changing the path to the basepoint only conjugates its image.*

Proof. Choose a spanning tree T_{Δ} of Δ and extend it to a spanning tree T of Γ . The edges outside T give the basis Σ , and those in $\Delta \setminus T_{\Delta}$ give the indicated subset. Collapsing T turns λ into a nonempty cyclically reduced word: cancellation between consecutive non-tree edges, including at the cyclic join, would require a closed tree segment between an edge and its inverse. A nonempty such segment cannot be reduced in a tree, while an empty segment would already be backtracking in λ .

Suppose that this word used only the letters from Δ . Between consecutive non-tree edges of λ , its tree segment is the unique tree path between vertices of Δ . That path lies in T_{Δ} . Every segment of λ would therefore lie in Δ , contrary to the hypothesis. The omitted-letter condition follows. The injection is the Freiheitssatz; see [18, Theorem 1]. $\square$

We use Linton's theorem in the following form: every Magnus subgroup of a hyperbolic one-relator group is quasiconvex [13, Theorem 4.7]. Together with Lemma 8.1, it applies to connected Magnus subgraphs of any graph presentation whose group has already been shown to be hyperbolic.

8.2 A finite domain in a cyclic cover

We use the geometric Magnus construction developed in [12, Section 4], giving the domain and complexity arguments in the form needed here. The construction finds a finite part of a cyclic cover whose adjacent overlaps will supply the two Magnus subgroups in an HNN splitting.

For a graph of rank at least two, choose an epimorphism

$$\phi: \pi_1 \Gamma \longrightarrow \mathbb{Z} \quad \text{with} \quad \phi([\lambda]) = 0.$$

Such a map exists: the abelianized relator imposes at most one equation on the integral characters of a free abelian group of rank at least two, and dividing a nonzero solution by the gcd of its coefficients makes it surjective. Let $p: \tilde{\Gamma} \rightarrow \Gamma$ be the corresponding connected cyclic cover, with deck transformation τ of height one. A *domain* will mean a finite connected subgraph $D \subset \tilde{\Gamma}$ such

that the occurrences of every base vertex and every base open edge form a nonempty interval of translates. Thus, after choosing any lift $\tilde{c}$ of a base cell c , the set

$$\{n \in \mathbb{Z} : \tau^n \tilde{c} \subset D\}$$

is a nonempty finite interval. In particular, the translates of a domain cover $\tilde{\Gamma}$. The following tree-overlap construction is the graph case of [12, Lemma 4.7].

Lemma 8.2. *Every connected cyclic cover of a finite connected graph has a domain D for which $D \cap \tau D$ is a nonempty tree.*

Proof. Choose a spanning tree downstairs. We may then index the lifted vertices by (v, n) so that tree edges preserve n , while each oriented non-tree edge has an integral increment d_e . These increments generate $\mathbb{Z}$. Let D_N be the subgraph induced by all vertex types at levels $[-N, N]$. For sufficiently large N , every base edge occurs and its occurrences form an interval.

We verify connectivity, including near the boundary of the interval. Choose a word in the signed increments whose sum is 1, and bound the absolute values of all its partial sums by L . Its translates connect consecutive levels whenever those levels are sufficiently far from the ends of $[-N, N]$; the lifted spanning trees connect the different vertex types at each level. Choose also a nonzero increment, with absolute value a . From a level near the lower boundary, repeated steps of size a in the positive direction reach the interior, and from the upper boundary repeated negative steps do the same. For $N > L + a$, these steps remain in the interval and reach the connected interior. Thus D_N is connected. The identical argument applies to the interval $[-N + 1, N]$, whose induced subgraph is $D_N \cap \tau D_N$.

Among domains with connected nonempty adjacent overlap, choose D with the fewest edges. If $D \cap \tau D$ contains a circuit, choose a simple one and translate it upward as long as it remains in D . Every position reached remains in $D \cap \tau D$, because its preceding translate lies in D . At the last position, some edge e of the circuit has $\tau e \notin D$. Since $e \in D \cap \tau D$, its preceding translate $\tau^{-1}e$ does lie in D . Delete the open edge e , retaining its endpoints. It was the highest occurrence of its base edge, so all occurrence intervals remain nonempty intervals. Moreover,

$$(D \setminus e) \cap \tau(D \setminus e) = (D \cap \tau D) \setminus e,$$

because $\tau e \notin D$. Both graphs remain connected: the deleted edge lies on a circuit in their common overlap. This contradicts minimality. The overlap is therefore a nonempty tree. $\square$

We next enlarge or modify such a domain to contain one lifted relator. Its two overlaps will become the edge groups of the splitting. The construction also accounts for strict decrease of the complexity

$$c(\Gamma, \lambda) = (|\lambda| - |V(S_\lambda)|, \text{rank } \pi_1 \Gamma), \quad (8.1)$$

ordered lexicographically. Here $|\lambda|$ counts edge traversals, with multiplicity. The first coordinate is nonnegative, since an immersed circuit visits at most one new vertex per edge traversal. Up to an additive constant in its second coordinate, this is the non-power form of the complexity used for one-relator hierarchies [12, Section 4.3].

Lemma 8.3. *Suppose $\text{rank } \pi_1 \Gamma \geq 2$ and that λ represents a non-power in $\pi_1 \Gamma$. There are a domain $U \subset \tilde{\Gamma}$ and a closed lift $\tilde{\lambda}: S^1 \rightarrow U$ with the following properties. Set*

$$U_- = U \cap \tau^{-1}U, \quad U_+ = U \cap \tau U, \quad Z = U \cup_{\tilde{\lambda}} D^2.$$

Both U_- and U_+ are connected nonempty Magnus subgraphs of Z . Either both are trees, or $U_- \cap U_+$ is connected and nonempty. Their fundamental groups are simultaneous Magnus subgroups in a presentation of $H = \pi_1 Z$, and

$$\pi_1 X \cong \langle H, t \mid tAt^{-1} = B \rangle, \quad A = \pi_1 U_-, \quad B = \pi_1 U_+, \quad (8.2)$$

with edge isomorphism induced by τ and suitable basepoint paths. The vertex and edge groups inject, the lifted relator is a non-power, and

$$c(U, \tilde{\lambda}) < c(\Gamma, \lambda).$$

Proof. We first construct U and prove the complexity decrease, using two cases according to the projection of the lifted support. We then verify the HNN splitting and the simultaneous Magnus bases for both constructions. This is the domain method of [12, Propositions 4.9 and 4.11].

Case 1: the lifted support does not project injectively on vertices. Take D from Lemma 8.2 and let

$$U = \bigcup_{i=0}^m \tau^i D,$$

where $m \geq 0$ is the least integer for which this union contains some closed lift $\tilde{\lambda}$. Such an integer exists: translate a fixed finite lifted circuit far enough in the positive direction, and then take a sufficiently long interval of translates of D . The graph U is connected and its cell occurrences are intervals. Indeed, an occurrence interval $[a, b]$ in D becomes $[a, b + m]$ in U . Figure 3 shows how taking adjacent overlaps trims the endpoints of this interval.

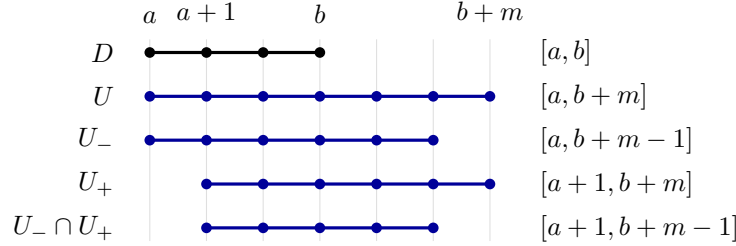


Figure 3: Occurrence intervals for one fixed base cell in Case 1, illustrated with $b - a = m = 3$. Each point represents a translate of that cell, not a copy of the domain D . The formulas also apply when $m = 0$, with an interval empty if its lower endpoint exceeds its upper endpoint. In that case $U = D$, and the complete adjacent overlap graphs are nonempty trees by Lemma 8.2.

If $m = 0$, both adjacent overlaps are trees. If $m \geq 1$, the interval calculation gives

$$U_- = \bigcup_{i=0}^{m-1} \tau^i D, \quad U_+ = \bigcup_{i=1}^m \tau^i D.$$

Minimality of m shows that neither contains a closed lift of λ . Their intersection is

$$U_- \cap U_+ = \begin{cases} D \cap \tau D, & m = 1, \\ \bigcup_{i=1}^{m-1} \tau^i D, & m \geq 2. \end{cases}$$

It is connected and nonempty. In the case $m = 0$, a tree cannot contain the nontrivial immersed circuit, so the overlaps are again Magnus subgraphs. The lifted circuit has the same length as λ and strictly more support vertices. Thus the first coordinate of (8.1) has decreased.

Case 2: the lifted support projects injectively on vertices. It then projects isomorphically onto S_λ : two lifts of the same oriented edge cannot have the same initial vertex. The character therefore vanishes on $\pi_1 S_\lambda$, and every lift of this support is a copy of it. Choose a spanning tree T_S of S_λ and extend it to a spanning tree of Γ ; in the resulting level coordinates each support copy lies at one level.

Contract the support downstairs and every support copy upstairs. This gives the connected cyclic cover of the quotient graph Γ/S_λ corresponding to the still-surjective character. Apply Lemma 8.2 there. Expand every occurrence of the contracted vertex in the resulting domain by a copy of T_S . At one chosen occurrence, add all the remaining edges of S_λ . Call the resulting graph U . It is a connected domain: the support-tree cells have the same occurrence interval as the contracted vertex, and each added support edge has a singleton occurrence interval. The overlap $U \cap \tau U$ is a tree, obtained by expanding vertices of a tree into trees; none of the added support edges lies in it. The full support at the chosen occurrence contains a closed lift $\tilde{\lambda}$. Both overlaps are trees and hence Magnus subgraphs.

Here the first complexity coordinate is unchanged and the second decreases. To see this directly, for each base open cell its number of occurrences in U exceeds its number of occurrences in $U \cap \tau U$ by one. Counting vertices and edges gives

$$\chi(U) - \chi(U \cap \tau U) = \chi(\Gamma).$$

Since the overlap is a tree, $\chi(U) = \chi(\Gamma) + 1$, and hence

$$\text{rank } \pi_1 U = \text{rank } \pi_1 \Gamma - 1.$$

The HNN splitting. Both constructions now have the required overlaps and complexity decrease. We recover X from the translates of Z , as in [12, Proposition 4.3]. Since $\phi([\lambda]) = 0$, the graph cover extends over the cells to the cyclic cover $Y \rightarrow X$. The translates $\tau^i Z$ cover Y , and their two-cell interiors are distinct. Form a line of disjoint copies of Z , joining each adjacent pair by the cylinder on its intersection. Denote this space by W ; collapsing these cylinders gives an equivariant map $W \rightarrow Y$. For a finite interval of copies, the intersection of a newly adjoined copy with the preceding union is precisely its adjacent overlap. Indeed, if a graph cell lies in the new copy and in an earlier copy, its occurrence interval also puts it in the immediately preceding copy. Distinct copies have no common two-cell interior, and cell boundaries consist of graph cells, so this checks the entire intersection. Van Kampen therefore shows inductively that the cylinder-collapse map induces a fundamental-group isomorphism over every finite interval. Every loop and every homotopy uses only finitely many copies, so $W \rightarrow Y$ induces an isomorphism on fundamental groups as well.

Taking the quotient by translation gives a graph of spaces with one vertex space Z and one edge space, and a map $W/\langle \tau \rangle \rightarrow X$. The two covering exact sequences have quotient $\mathbb{Z}$; the map is an isomorphism on their kernels and on this quotient. It follows that it induces an isomorphism on fundamental groups. The edge maps are injective by Lemma 8.1, because each overlap omits a traversed relator edge. Thus this graph of spaces gives the HNN extension (8.2), and the normal form theorem gives injection of H into $\pi_1 X$.

Simultaneous Magnus bases and the edge isomorphism. It remains to choose the two Magnus subsets simultaneously and specify their basepoint paths. If the overlap groups are trivial, this is

immediate. Otherwise put $U_0 = U_- \cap U_+$, which is connected and nonempty by the construction. Extend a spanning tree of U_0 to spanning trees of U_- and U_+ . Their intersection is exactly the chosen tree of U_0 , so their union is a tree. Extend that union to a spanning tree of U . With basepoint in U_0 , the non-tree edges of the two overlaps are then subsets of a single free basis of $\pi_1 U$. Their common letters freely generate $C = \pi_1 U_0$, which is a free factor in each side group.

To specify the stable letter in this nontrivial-edge case, choose $x \in U_0$ and a path $q \subset U_+$ from x to τx . The latter vertex belongs to U_+ because $x \in U_-$. Then $t = [p(q)]$ has height one, and the edge isomorphism sends a loop a in U_- to $q(\tau a)q^{-1}$ in U_+ . Its projection is tat^{-1} , giving the specified isomorphism in (8.2). When the overlap groups are trivial, choose instead any $x \in U_-$ and a path $q \subset U$ from x to τx . Again $t = [p(q)]$ has height one, and the edge relation is empty. In either case, q in the copy Z closes in the quotient graph of spaces by crossing the cylinder to the copy τZ at τx . It crosses the quotient loop edge exactly once, so it is a stable letter; collapsing the cylinder sends it to $p(q)$.

Finally, $\pi_1 U$ injects into $\pi_1 \Gamma$, by inclusion into the graph cover and its covering map. If the lifted relator were a proper power in $\pi_1 U$, its image would be a proper power in $\pi_1 \Gamma$. This proves the remaining assertion. $\square$

8.3 Hyperbolicity and the final reductions

The preceding construction supplies the geometric input to the combination theorem. We now check its algebraic hypotheses.

Proposition 8.4. *Let Γ be a finite connected graph, and let $\lambda: S^1 \rightarrow \Gamma$ be a nontrivial immersed circuit representing a non-power in $\pi_1 \Gamma$. If $\pi_1(\Gamma \cup_\lambda D^2)$ is BS -free, then it is word-hyperbolic.*

Proof. We induct on the nonnegative lexicographic complexity (8.1). In graph rank one, the non-power relator generates the infinite cyclic fundamental group, and the quotient is trivial. Rank zero cannot support the stipulated circuit.

In higher rank, apply Lemma 8.3 to obtain

$$G = \langle H, t \mid tAt^{-1} = B \rangle.$$

The vertex group H embeds in G , and so is BS -free. It has a non-power graph presentation of smaller complexity, hence is hyperbolic by induction. The one-relator torsion theorem makes it torsion-free [18, Theorem 2]. The side groups A and B are finitely generated free Magnus subgroups of H ; by [13, Theorem 4.7], they are quasiconvex in H .

Use the simultaneous bases from Lemma 8.3. Collins's conjugate-intersection theorem [4, Theorem 2], which permits equal Magnus subgroups, says that for $M, N \in \{A, B\}$ and $g \in H$,

$$M \cap gNg^{-1} \text{ noncyclic} \implies g \in MN.$$

This is the first overlap hypothesis of Theorem 3.1. Collins's direct-intersection theorem [4, Theorem 1] gives

$$I = A \cap B = C \quad \text{or} \quad I = C * \langle u \rangle,$$

where C is generated by the common basis letters and the second expression is an internal free product with an infinite cyclic group. Thus I and its transported copy $J = t^{-1}It$ are finitely generated. If the edge group is nontrivial, the two Magnus subgraphs U_-, U_+ have connected intersection by the construction. Linton's strong-inertia lemma [12, Lemma 2.10] states that the intersection of the fundamental groups of two such Magnus subgraphs is strongly inert in

either side group. It follows that I is strongly inert in $P = A$ and in B . Isomorphisms preserve reduced ranks and the indexing double cosets, so transporting the latter assertion through the edge isomorphism makes J strongly inert in P as well. These are precisely the two strong reduced-rank inertia inequalities in Theorem 3.1. When the edge group is trivial, the same requirements hold directly.

All hypotheses of Theorem 3.1 have now been verified. Its conclusion makes G hyperbolic, completing the induction. $\square$

Proof of Theorem 1.1. Freely and cyclically reduce the defining relator; conjugating or inverting it does not change its normal closure. If it becomes trivial, the group is a finitely generated free group and is hyperbolic.

If the cyclically reduced relator is a proper power, write it as w^k , where w is a nontrivial cyclically reduced non-power and $k \geq 2$. Newman's spelling theorem [20], in the form [18, Theorem 4], says that every nonempty freely reduced null word contains a segment of a cyclic conjugate of $w^{\pm k}$ of length greater than $(k-1)|w|$. This is strictly more than half the relator length, including when $k = 2$. Replacing that segment by the inverse complementary segment strictly shortens the word and uses one defining relation. Repeating, with free reduction after each replacement, fills a null word with at most its original length in relator cells. These shortening rules give a finite Dehn presentation (and a linear isoperimetric inequality). The group is therefore word-hyperbolic [2, Theorem III.Γ.2.6].

In the remaining case, realize the cyclically reduced non-power relator as an immersed circuit in the basis rose. Proposition 8.4 applies and proves hyperbolicity under the stated exclusion of $BS(m, n)$ for all nonzero integers m, n . $\square$

Proof of Theorem 1.2. Under its non-power graph-presentation hypotheses, Proposition 8.4 makes the ambient group hyperbolic. For any connected subgraph omitting a traversed edge, Lemma 8.1 identifies its fundamental group with an injective Magnus subgroup in a one-relator presentation. Linton's theorem [13, Theorem 4.7] makes that subgroup quasiconvex. Basepoint paths only conjugate it, so the assertion holds for every choice of such paths. $\square$

9 Virtual free-by-cyclic consequence

We now combine the hyperbolicity theorem with the companion virtual compact specialness theorem and the results of Kielak–Linton.

Corollary 9.1. *Let X be finite, let $r \in F(X)$, and suppose that $G = F(X)/\langle\langle r \rangle\rangle$ contains no subgroup isomorphic to $BS(m, n)$ for any nonzero integers m, n . There are a finite-index subgroup $H \leq G$, an integer $d \geq 0$, an automorphism α of the rank- d free group F_d , and an embedding*

$$H \hookrightarrow F_d \rtimes_{\alpha} \mathbb{Z}.$$

Consequently the same H fits into an exact sequence

$$1 \longrightarrow K \longrightarrow H \longrightarrow Q \longrightarrow 1, \quad K \text{ free}, \quad Q \leq \mathbb{Z}.$$

Thus G is virtually free-by-cyclic, with the rank of K allowed to be infinite and Q allowed to be trivial.

Proof. If G is finite, take $H = 1$ and $F_d = F_0 = 1$, with ambient group $\mathbb{Z}$. Otherwise Theorem 1.1 makes G hyperbolic, and Theorem 1.1 of [21] makes it virtually compact special. The free group $F(X)$ is the fundamental group of a compact connected planar surface with $|X| + 1$ boundary components. Kielak–Linton’s Corollary 1.3 [10] therefore makes G virtually free-by-cyclic. The surface may be a disk, and their relator is arbitrary; trivial relators and proper powers are not excluded.

For a hyperbolic virtually compact special group, Theorem 1.1 of [10] equates being virtually free-by-cyclic with a finite-index embedding in a free-by-cyclic group with finitely generated free kernel. In the rational cohomological dimension two branch of their proof the ambient quotient is $\mathbb{Z}$. In the dimension-at-most-one branch they obtain a finite-index free subgroup, which has finite rank here and embeds in its direct product with $\mathbb{Z}$. In either branch the ambient extension splits over $\mathbb{Z}$, giving the displayed semidirect product.

Identify H with its image and restrict the projection onto $\mathbb{Z}$. Its kernel is $K = H \cap F_d$, which is free as a subgroup of a free group, and its image is a subgroup $Q \leq \mathbb{Z}$. This is the asserted exact sequence. Finite generation of the ambient kernel F_d does not imply finite generation of K . $\square$

The companion also proves that

$$F*_\phi = \langle F, t \mid t^{-1}ft = \phi(f) \ (f \in F) \rangle$$

is virtually compact special whenever F is free of finite rank, $\phi : F \rightarrow F$ is injective, and $F*_\phi$ is word-hyperbolic [21, Theorem 1.2]. Such a group is finitely generated, so Kielak–Linton’s Corollary 1.8 [10], followed by the same ambient argument, gives the finite-index embedding and subgroup-closed free-by-cyclic conclusion for these ascending HNN extensions as well. The resulting free kernel is not asserted to be their canonical height kernel.

References

- [1] Mladen Bestvina and Mark Feighn. Addendum and correction to “a combination theorem for negatively curved groups”. *Journal of Differential Geometry*, 43(4):783–788, 1996. doi:10.4310/jdg/1214458531.
- [2] Martin R. Bridson and André Haefliger. *Metric Spaces of Non-Positive Curvature*, volume 319 of *Grundlehren der mathematischen Wissenschaften*. Springer, 1999. doi:10.1007/978-3-662-12494-9.
- [3] Peter Brinkmann. Hyperbolic automorphisms of free groups. *Geometric and Functional Analysis*, 10(5):1071–1089, 2000. Citations refer to arXiv:math/9906008v3, 28 September 1999; doi:10.1007/PL00001647.
- [4] Donald J. Collins. Intersections of conjugates of Magnus subgroups of one-relator groups. *Geometry & Topology Monographs*, 14:135–171, 2008. doi:10.2140/gtm.2008.14.135.
- [5] Giles Gardam, Dawid Kielak, and Alan D. Logan. Algebraically hyperbolic groups. *Groups, Geometry, and Dynamics*, 2025. doi:10.4171/GGD/907.
- [6] S. M. Gersten. Problems on automatic groups. In Gilbert Baumslag and Charles F. Miller, III, editors, *Algorithms and Classification in Combinatorial Group Theory*, volume 23 of *Mathematical Sciences Research Institute Publications*, pages 225–232. Springer, New York, 1992. doi:10.1007/978-1-4613-9730-4_10.

- [7] Rita Gitik, Mahan Mitra, Eliyahu Rips, and Michah Sageev. Widths of subgroups. *Transactions of the American Mathematical Society*, 350(1):321–329, 1998. doi:[10.1090/S0002-9947-98-01792-9](https://doi.org/10.1090/S0002-9947-98-01792-9).
- [8] Ilya Kapovich. A non-quasiconvexity embedding theorem for hyperbolic groups. *Mathematical Proceedings of the Cambridge Philosophical Society*, 127(3):461–486, 1999. Citations refer to [arXiv:math/9704203v1](https://arxiv.org/abs/math/9704203v1), 11 April 1997; doi:[10.1017/S0305004199003862](https://doi.org/10.1017/S0305004199003862).
- [9] Ilya Kapovich. Currents on free groups. In *Topological and Asymptotic Aspects of Group Theory*, volume 394 of *Contemporary Mathematics*, pages 149–176. American Mathematical Society, 2006. doi:[10.1090/conm/394/07441](https://doi.org/10.1090/conm/394/07441).
- [10] Dawid Kielak and Marco Linton. Virtually free-by-cyclic groups. *Geometric and Functional Analysis*, 34(5):1580–1608, 2024. doi:[10.1007/s00039-024-00687-6](https://doi.org/10.1007/s00039-024-00687-6).
- [11] Marco Linton. The geometry of subgroups of mapping tori of free groups, 2025. Citations refer to [arXiv:2510.03145v1](https://arxiv.org/abs/2510.03145v1), 3 October 2025.
- [12] Marco Linton. One-relator hierarchies. *Duke Mathematical Journal*, 174(4):747–802, 2025. Citations refer to [arXiv:2202.11324v3](https://arxiv.org/abs/2202.11324v3), 27 June 2024; doi:[10.1215/00127094-2024-0040](https://doi.org/10.1215/00127094-2024-0040).
- [13] Marco Linton. Hyperbolic one-relator groups. *Canadian Journal of Mathematics*, 78(1):35–61, 2026. doi:[10.4153/S0008414X24000427](https://doi.org/10.4153/S0008414X24000427).
- [14] Marco Linton and Carl-Fredrik Nyberg-Brodda. The theory of one-relator groups: history and recent progress, 2025. [arXiv:2501.18306v1](https://arxiv.org/abs/2501.18306v1), 30 January 2025.
- [15] Larsen Louder and Henry Wilton. Negative immersions for one-relator groups. *Duke Mathematical Journal*, 171(3):547–594, 2022. Citations refer to [arXiv:1803.02671v4](https://arxiv.org/abs/1803.02671v4), 6 May 2021; doi:[10.1215/00127094-2021-0024](https://doi.org/10.1215/00127094-2021-0024).
- [16] Larsen Louder and Henry Wilton. Uniform negative immersions and the coherence of one-relator groups. *Inventiones mathematicae*, 236(2):673–712, 2024. doi:[10.1007/s00222-024-01246-4](https://doi.org/10.1007/s00222-024-01246-4).
- [17] Wilhelm Magnus. Über diskontinuierliche Gruppen mit einer definierenden Relation. (Der Freiheitssatz). *Journal für die reine und angewandte Mathematik*, 163:141–165, 1930. doi:[10.1515/crll.1930.163.141](https://doi.org/10.1515/crll.1930.163.141).
- [18] James McCool and Paul E. Schupp. On one relator groups and HNN extensions. *Journal of the Australian Mathematical Society*, 16(2):249–256, 1973. doi:[10.1017/S1446788700014300](https://doi.org/10.1017/S1446788700014300).
- [19] Jean Pierre Mutanguha. The dynamics and geometry of free group endomorphisms. *Advances in Mathematics*, 384:107714, 2021. Citations refer to [arXiv:2005.11896v2](https://arxiv.org/abs/2005.11896v2), 6 January 2021; doi:[10.1016/j.aim.2021.107714](https://doi.org/10.1016/j.aim.2021.107714).
- [20] B. B. Newman. Some results on one-relator groups. *Bulletin of the American Mathematical Society*, 74:568–571, 1968. doi:[10.1090/S0002-9904-1968-12012-9](https://doi.org/10.1090/S0002-9904-1968-12012-9).
- [21] OpenAI. Virtual compact specialness of hyperbolic one-relator groups. OpenAI Math Release preprint [OAI:Virtual-compact-specialness-of-hyperbolic-one-relator-groups-September-25-2026](https://arxiv.org/abs/2509.12345), 2026.

- [22] John R. Stallings. Topology of finite graphs. *Inventiones mathematicae*, 71(3):551–565, 1983.
[doi:10.1007/BF02095993](https://doi.org/10.1007/BF02095993).